THE STACKCALC LIBRARY • CURRICULUM SERIES — BOOK 4 OF 4

STACK SPEED & FLUENCY

15 Speed Lessons, Shortcut Playbooks & 60 Progressive Practice Tests (1,200 Problems)

A complete speed-training and mathematical fluency coaching book for homeschooling families, classrooms, and competitive math/calculator teams. Every section opens with a 1-Page Speed Lesson & Shortcut Playbook ('Slow Way vs. Champion Way' worked archetypes plus 6 guided warm-up drills), followed by four 20-problem progressive practice tests (Tests A–D) and complete answer keys with optimal Champion Pipelines.

StackCalc

stackcalc.io

First Edition

BOOKS, BLOCKS & INSTRUMENTS IN THE STACKCALC LIBRARY

Illustrated Companion Trade Books (History of Mathematics & Stack Thinking)

- Volume I — From Blocks to the Stack: How Hands-On Blocks Make Numbers Click (Grades K–6 / StackCalc SC-6)
- Volume II — Thinking in the Stack: The Art of Calculating Without Parentheses (Grades 7–12, AP & Engineers / StackCalc SC-32)

K–12 Concept-First Curriculum & Student Workbooks (US Letter 8.5" × 11" Editions)

- Book 1 — Teaching with the Stack: Unified K–12 Master Teacher's Guide, Concept Lessons & Complete Answer Keys
- Book 2 — Hands-On Math Lab: K–12 Student Manipulative & Four-Line Stack Exploration Workbook
- Book 3 — Middle School Math Lab: Grades 6–8 Two-Method Student Practice Workbook (900 Problems) & 15-Page Coach Guide
- Book 4 — Stack Speed & Fluency: 15 Speed Lessons, Shortcut Playbooks & 60 Progressive Practice Tests (1,200 Problems)

Three Published Physical Manipulative Block Studios & Four-Line Stack Calculators (stackcalc.io)

- Place Value Studio: 6×2 Whole & Decimal Trays, Digit Blocks (0–9) & Stackable Gold [+1] Carry Tokens
- Fraction Studio: 84.0 mm Proportional Fraction Towers (1 Whole to 1/12) & 12-Socket Comparator Lid
- Prime Factor Studio: Logarithmic Prime Blocks (2, 3, 5, 7 where equal products stand flush) & GCF/LCM Comparator
- StackCalc Calculators: SC-6 Elementary/Middle School Calculator & SC-32 Scientific Calculator (Hardware, iOS, iPadOS, watchOS & Web)

COPYRIGHT, STATE STANDARDS ALIGNMENT & OPEN-EDUCATION LICENSE

STACK SPEED & FLUENCY: 15 Speed Lessons, Shortcut Playbooks & 60 Progressive Practice Tests (1,200 Problems)

Copyright © 2026 by StackCalc (https://stackcalc.io). Some rights reserved. Published by StackCalc Learning Lab, San Francisco, California. First Edition (US Letter 8.50 × 11.00 in / 215.9 × 279.4 mm).

State & District Curriculum Alignment:

Designed to supplement state-mandated mathematics standards—including Common Core State Standards (CCSS-M), Texas Essential Knowledge and Skills (TEKS), Florida B.E.S.T., Virginia SOL, and AP Mathematics / Computer Science—alongside district basal textbooks (Eureka Math / EngageNY, Illustrative Mathematics / Open Up, enVision Mathematics, Big Ideas Math, HMH Into Math, McGraw-Hill Reveal, Bridges in Mathematics, and Singapore Dimensions Math).

Open-Education License (CC BY-NC-SA 4.0) & Perpetual Classroom Reproduction Permission:

Classroom teachers, math coaches, interventionists, homeschool families, and schools are expressly permitted to photocopy or print student lab worksheets, speed trials, reference cards, and parent welcome letters for non-commercial educational instruction under Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International.

Regulatory & Physical Safety Notice (Reese's Law / P.L. 117-171 / 16 CFR Part 1263 & ASTM F963):

Physical calculator instruments that utilize coin or button cell batteries must maintain a captive, tool-secured battery compartment compliant with UL 4200A. 3D-printed manipulative blocks constitute small parts under 16 CFR Part 1501; adult supervision is required for children under 6 years of age.

Trademark Notice & Print Colophon:

StackCalc, SC-32, and SC-6 are trademarks of StackCalc. All curriculum and textbook names are trademarks of their respective owners, none of whom endorse or sponsor this independent publication. Typeset in Computer Modern Unicode (CMU Serif), SpaceMono, and STIX Math in high-contrast monochrome ink on archival white paper.

Table of Contents

Directory of the Preface, Master Score Tracker, 15 Speed Lessons + Test Sections (Tests A–D), and Complete Answer Keys.

Section / Speed Lesson & Practice Test Series	Lesson & Tests Included	Difficulty Progression	Pages
Preface: The 3-Step Speed Training Cycle	Students, Families & Teams	1. Lesson → 2. Drill → 3. Tests	**p. iv**
Master Score & Personal-Best Progress Tracker	All 60 Practice Tests	Score Log Across Levels 1 to 4	**p. 1**
Section 1: Step Counting, Benchmarks & Sequences	Lesson + Tests 1A–1D (86 probs)	Level 1 → Level 4 (Foundational to Advanced)	**pp. 2–4**
Section 2: Number Bonds & Missing Addends	Lesson + Tests 2A–2D (86 probs)	Level 1 → Level 4 (Foundational to Advanced)	**pp. 5–7**
Section 3: Complements & Make-10 Bridging	Lesson + Tests 3A–3D (86 probs)	Level 1 → Level 4 (Foundational to Advanced)	**pp. 8–10**
Section 4: Doubles, Equal Groups & Squares	Lesson + Tests 4A–4D (86 probs)	Level 1 → Level 4 (Foundational to Advanced)	**pp. 11–13**
Cumulative Review C1: Number Sense (Sections 1–4)	Lesson + Tests C1A–C1D (86 probs)	Level 1 → Level 4 (Foundational to Advanced)	**pp. 14–16**
Section 5: Multi-Step Operations & Grouping	Lesson + Tests 5A–5D (86 probs)	Level 1 → Level 4 (Grades 3–8+)	**pp. 17–19**
Section 6: Fractions, Decimals & Reciprocals	Lesson + Tests 6A–6D (86 probs)	Level 1 → Level 4 (Grades 4–9+)	**pp. 20–22**
Section 7: Primes, GCF/LCM & Nested Expressions	Lesson + Tests 7A–7D (86 probs)	Level 1 → Level 4 (Grades 5–10+)	**pp. 23–25**
Section 8: Algebra, Quadratics & Coordinate Geometry	Lesson + Tests 8A–8D (86 probs)	Level 1 → Level 4 (Grades 6–10+)	**pp. 26–28**
Cumulative Review C2: Intermediate Math (Sections 1–8)	Lesson + Tests C2A–C2D (86 probs)	Level 1 → Level 4 (Grades 5–10+)	**pp. 29–31**
Section 9: Trigonometry, Vectors & Polar Coordinates	Lesson + Tests 9A–9D (86 probs)	Level 1 → Level 4 (Grades 8–12+)	**pp. 32–34**
Section 10: Exponents, Logarithms & Growth Models	Lesson + Tests 10A–10D (86 probs)	Level 1 → Level 4 (Grades 8–12+)	**pp. 35–37**
Section 11: Combinatorics, Probability & Statistics	Lesson + Tests 11A–11D (86 probs)	Level 1 → Level 4 (Grades 7–12+)	**pp. 38–40**
Section 12: Base Conversions, Binary & Bitwise Logic	Lesson + Tests 12A–12D (86 probs)	Level 1 → Level 4 (Grades 7–12+)	**pp. 41–43**
Cumulative Review C3: All-Level Championship (Sections 1–12)	Lesson + Tests C3A–C3D (86 probs)	Level 1 → Level 4 (Grades 7–12+)	**pp. 44–46**
Appendix A: Complete Answer Keys & Champion Pipelines	All 60 Tests + Pipelines	Answers & Optimal Stack Routes	**pp. 47–51**
Appendix B: Mental Math & Stack Speed Tactics	SC-6 & SC-32	Five Inside-Out Verification Patterns	**p. 52**
Appendix C: Progress Graph & Mastery Certificate	Personal Best Log	Score Progression Chart & Certificate	**p. 53**

Preface: How to Train for Mathematical Speed & Fluency

Why raw test pages alone do not build speed—and how each 3-page section trains champion habits step by step.

---

1. THE 3-PAGE ARCHITECTURE OF EVERY SECTION (LESSON → DRILL → TESTS A–D)

Taking timed tests over and over without learning faster methods only makes slow habits permanent. That is why every section in this book is organized as a complete 3-page training module:

- Page 1 — Speed Lesson, 'Slow Way vs. Champion Way' Playbook & 6 Guided Warm-Up Drills: Teaches the exact algebraic identities, mental shortcuts, and zero-memory stack pipelines before the clock starts.
- Page 2 — Progressive Tests A & B (Levels 1 & 2, 40 problems): Builds foundation recall and core fluency.
- Page 3 — Progressive Tests C & D (Levels 3 & 4, 40 problems): Reaches multi-step speed and tournament challenge.

2. THREE WAYS TO USE THIS BOOK AT HOME, IN CLASS, OR ON A COMPETITIVE TEAM

Mode 1 — Structured Lesson + Untimed Mastery (Start at Level 1, Grow to Level 4):

Read the 3 Speed Rules and 'Slow Way vs. Champion Way' boxes on the first page of the section, complete the 6 Guided Warm-Up problems untimed, and then work through Test A (Level 1) and Test B (Level 2) at a comfortable pace.

Champion Pipeline: 2 4 ENTER 1 8 +

Mode 2 — Friday Curriculum Fluency Companion (Works Alongside Any Math Program):

Pair any section with Singapore Math, Saxon, RightStart, Math Mammoth, AoPS, or school textbooks. Students write their answers on the clean test lines, then verify on their StackCalc calculator or check Appendix A (pp. 47–51).

Champion Pipeline: 1 4 ENTER 6 + 9 ENTER 4 - ÷

Mode 3 — Competitive Calculator & Number Sense Team Training (Tests C & D + Error Audit):

Set a timer for the Goal at the top of the test strip (3 to 6 minutes). After grading in Appendix A, compare your keystrokes against the Optimal Champion Pipelines printed in every section's answer key box.

Champion Pipeline: 4 x^{-1} 6 x^{-1} + x^{-1}

3. SIMPLE FLUENCY SCORING OR CONTEST SCORING (20 PROBLEMS PER TEST)

- Standard Fluency Score (Out of 20): Count 1 point for each correct answer (Score: _____ / 20), or multiply by 5 for a percentage score (18 / 20 = 90%).
- Competitive Team Scoring: Award +5 points per correct answer and deduct 2 points per incorrect answer (max score 100 points per 20-problem test). Never guess wildly—use magnitude estimation to catch errors before writing!

MASTER SCORE & PERSONAL-BEST PROGRESS TRACKER • ALL 15 SECTIONS & 60 TESTS

Speed & Fluency Progress Tracker

Check off each section's Warm-Up Drill and record your score (out of 20) and time across Levels 1 to 4.

HOW TO LOG YOUR PROGRESS ACROSS LESSONS, WARM-UPS & LEVELS 1 TO 4

Start every section by reading the Speed Lesson & Shortcut Playbook and solving the 6 Warm-Up Drills on the first page of the section. Then record your score (___/20) and best completion time across Tests A, B, C, and D!

Sec	Mathematical Topic	Test A (Lvl 1)	Test B (Lvl 2)	Test C (Lvl 3)	Test D (Lvl 4)	Pages
1	**Step Counting, Benchmarks & Sequences** Foundational to Advanced • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	2–4
2	**Number Bonds & Missing Addends** Foundational to Advanced • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	5–7
3	**Complements & Make-10 Bridging** Foundational to Advanced • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	8–10
4	**Doubles, Equal Groups & Squares** Foundational to Advanced • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	11–13
C1	**Number Sense (Sections 1–4)** Foundational to Advanced • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	14–16
5	**Multi-Step Operations & Grouping** Grades 3–8+ • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	17–19
6	**Fractions, Decimals & Reciprocals** Grades 4–9+ • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	20–22
7	**Primes, GCF/LCM & Nested Expressions** Grades 5–10+ • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	23–25
8	**Algebra, Quadratics & Coordinate Geometry** Grades 6–10+ • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	26–28
C2	**Intermediate Math (Sections 1–8)** Grades 5–10+ • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	29–31
9	**Trigonometry, Vectors & Polar Coordinates** Grades 8–12+ • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	32–34
10	**Exponents, Logarithms & Growth Models** Grades 8–12+ • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	35–37
11	**Combinatorics, Probability & Statistics** Grades 7–12+ • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	38–40
12	**Base Conversions, Binary & Bitwise Logic** Grades 7–12+ • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	41–43
C3	**All-Level Championship (Sections 1–12)** Grades 7–12+ • [ ] Lesson & Drill	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	Score: ___/20 Time: _____	44–46

Checking Your Answers & Optimal Champion Pipelines

Turn to Appendix A (Pages 47–51) for the complete 60-test answer key and Optimal Champion Pipelines showing the fastest keystroke/mental route for every section's challenge problems.

Section 1: Step Counting, Benchmarks & Sequences

Unit steps, skip-counting, 10/100 benchmarks, and arithmetic series. • Study the Champion Way & complete W1–W6 before starting Tests 1A–1D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 1: STEP COUNTING, BENCHMARKS & SEQUENCES

RULE 1	**Rule 1 — Pair Ends to Land on 10s & 100s First: In multi-step jumps (e.g., 68 + 12 + 25), combine the pair that completes a full 10 or 100 first (68 + 12 = 80, then + 25 = 105).**
RULE 2	Rule 2 — Gauss Pairing for Arithmetic Series: To sum an evenly spaced sequence (First + ... + Last), multiply the average of the first and last terms by the number of terms: Sum = n × (First + Last) ÷ 2.
RULE 3	Rule 3 — Sum of First n Odd Numbers Is n^2: Never add 1 + 3 + 5 + ... one by one! Count how many odd numbers there are and square the count (first 5 odds = $5^2 = 25$; first 6 odds = $6^2 = 36$).

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — Benchmark Pairing: Evaluate 96 + 4 + 38

Stack Keys: `9` `6` `ENTER` `4` `+` `3` `8` `+`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Add digits column by column on paper (6+4+8 = 18, carry 1).

CHAMPION SPEED WAY (Use This!):

Champion Way: Spot that 96 + 4 = 100 immediately, then 100 + 38 = 138 in 1 second!

Archetype 2 — Gauss Rainbow Pairing: Evaluate 14 + 16 + 18 + 20 + 22

Stack Keys: `1` `8` `ENTER` `5` `×`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Add left-to-right (14+16=30, +18=48, +20=68, +22=90) — 4 separate additions.

CHAMPION SPEED WAY (Use This!):

Champion Way: For any symmetric 5-term sequence, the sum is simply 5 × Middle Term: 5 × 18 = 90!

Archetype 3 — Triangular Series Sum: Evaluate 1 + 2 + 3 + ... + 10

Stack Keys: `1` `0` `ENTER` `1` `1` `×` `2` `÷`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Type all 10 numbers into the calculator one by one (29 keystrokes).

CHAMPION SPEED WAY (Use This!):

Champion Way: Use n(n + 1) ÷ 2: 10 × 11 ÷ 2 = 55 in 1 mental step (or 7 keys on the stack)!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. $76 + 4 + 19 =$ _____

Shortcut Cue: Complete 80 first, then + 19

W2. $150 - 25 - 25 =$ _____

Shortcut Cue: Combine 25 + 25 = 50, subtract from 150

W3. $1 + 3 + 5 + 7 =$ _____

Shortcut Cue: 4 consecutive odd numbers → 4^2

W4. $10 + 15 + 20 + 25 + 30 =$ _____

Shortcut Cue: 5 terms × middle term (20)

W5. $1 + 2 + 3 + \ldots + 8 =$ _____

Shortcut Cue: Gauss rule: 8 × 9 ÷ 2

W6. $2 + 4 + 6 + 8 + 10 =$ _____

Shortcut Cue: n(n + 1) for 5 even terms: 5 × 6

Warm-Up Self-Check Answers: W1: 99 | W2: 100 | W3: 16 | W4: 100 | W5: 36 | W6: 30

Section 1: Step Counting, Benchmarks & Sequences

Unit steps, skip-counting, 10/100 benchmarks, and arithmetic series. • Foundation (Test 1A) & Core Fluency (Test 1B)

TEST 1A • LEVEL 1: FOUNDATION Goal: 3 min

Adding & Subtracting 1, 2, and 3 Within 20

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1.	3 + 1 =	
2.	5 + 1 =	
3.	8 + 1 =	
4.	6 - 1 =	
5.	10 - 1 =	
6.	4 + 2 =	
7.	7 + 2 =	
8.	9 - 2 =	
9.	2 + 1 + 1 =	
10.	5 + 1 + 1 =	
11.	8 - 1 - 1 =	
12.	11 + 1 =	
13.	14 - 1 =	
14.	12 + 2 =	
15.	16 - 2 =	
16.	10 + 3 =	
17.	15 - 3 =	
18.	7 + 1 + 2 =	
19.	13 + 2 + 1 =	
20.	19 - 2 - 1 =	

TEST 1B • LEVEL 2: CORE FLUENCY Goal: 4 min

Skip-Counting Steps by 2s, 5s, and 10s

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1.	2 + 2 + 2 =	
2.	2 + 2 + 2 + 2 =	
3.	5 + 5 =	
4.	5 + 5 + 5 =	
5.	10 + 10 + 10 =	
6.	12 + 2 + 2 =	
7.	18 - 2 - 2 =	
8.	15 + 5 + 5 =	
9.	25 - 5 - 5 =	
10.	20 + 10 + 5 =	
11.	30 - 10 - 5 =	
12.	14 + 3 + 3 =	
13.	22 + 4 + 4 =	
14.	40 - 5 - 5 =	
15.	35 + 10 + 5 =	
16.	50 - 10 - 10 =	
17.	16 + 2 + 2 + 2 =	
18.	45 + 5 + 10 =	
19.	70 - 10 - 5 =	
20.	85 + 5 + 10 =	

Section 1: Step Counting, Benchmarks & Sequences

Unit steps, skip-counting, 10/100 benchmarks, and arithmetic series. • Multi-Step Speed (Test 1C) & Advanced Challenge (Test 1D)

TEST 1C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Crossing 10s and 100s with Multi-Step Jumps

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	28 + 2 + 6 =	_____
2.	37 + 3 + 8 =	_____
3.	54 - 4 - 7 =	_____
4.	63 - 3 - 9 =	_____
5.	45 + 15 + 10 =	_____
6.	82 - 12 - 15 =	_____
7.	96 + 4 + 25 =	_____
8.	115 - 15 - 20 =	_____
9.	75 + 25 + 35 =	_____
10.	150 - 25 - 25 =	_____
11.	68 + 12 + 20 =	_____
12.	134 - 14 - 30 =	_____
13.	180 + 20 + 45 =	_____
14.	260 - 60 - 35 =	_____
15.	125 + 25 + 50 =	_____
16.	310 - 10 - 45 =	_____
17.	240 + 30 + 30 =	_____
18.	420 - 70 - 50 =	_____
19.	375 + 25 + 80 =	_____
20.	500 - 125 - 75 =	_____

TEST 1D • LEVEL 4: ADVANCED CHALLENGE Goal: 5 min

Arithmetic Sequences & Fast Series Sums

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	1 + 2 + 3 + 4 + 5 =	_____
2.	2 + 4 + 6 + 8 + 10 =	_____
3.	1 + 3 + 5 + 7 + 9 =	_____
4.	3 + 6 + 9 + 12 =	_____
5.	4 + 8 + 12 + 16 =	_____
6.	5 + 10 + 15 + 20 + 25 =	_____
7.	1 + 2 + 3 + 4 + 5 + 6 + 7 =	_____
8.	10 + 12 + 14 + 16 =	_____
9.	15 + 20 + 25 + 30 =	_____
10.	11 + 13 + 15 + 17 =	_____
11.	20 + 25 + 30 + 35 + 40 =	_____
12.	1 + 3 + 5 + 7 + 9 + 11 =	_____
13.	6 + 12 + 18 + 24 =	_____
14.	100 - 12 - 18 - 20 =	_____
15.	25 + 50 + 75 + 100 =	_____
16.	14 + 16 + 18 + 20 + 22 =	_____
17.	7 + 14 + 21 + 28 =	_____
18.	50 + 60 + 70 + 80 + 90 =	_____
19.	1 + 2 + 3 + ... + 10 =	_____
20.	2 + 4 + 6 + ... + 20 =	_____

Section 2: Number Bonds & Missing Addends

Part-part-whole composition, decomposition, and missing-part equations. • Study the Champion Way & complete W1–W6 before starting Tests 2A–2D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 2: NUMBER BONDS & MISSING ADDENDS

RULE 1	**Rule 1 — Think Addition to Solve Missing Addends: For 28 + [?] = 50, count up to the next ten (+2 → 30) and then to 50 (+20) → 22, or key [50] [ENTER] [28] [-].**
RULE 2	Rule 2 — Subtract in Two Clean Chunks (Tens Then Ones): For 72 - 35, subtract 30 first (72 - 30 = 42), then subtract 5 (42 - 5 = 37).
RULE 3	Rule 3 — Balanced Equation Shifting: For 85 + 45 = 100 + [?], notice 100 is 15 more than 85, so the missing number is 15 less than 45 (45 - 15 = 30)!

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — Missing Addend to 100: Solve 63 + [?] = 100 — Stack Keys: `1` `0` `0` `ENTER` `6` `3` `-`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Write vertical 100 - 63 on paper and borrow across two zeros.

CHAMPION SPEED WAY (Use This!):

Champion Way: Make 9 tens and 10 ones: (9 - 6 = 3 tens) and (10 - 3 = 7 ones) → 37!

Archetype 2 — Reordering Three-Part Sums: Evaluate 48 + 35 + 52 — Stack Keys: `4` `8` `ENTER` `5` `2` `+` `3` `5` `+`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Compute 48 + 35 = 83 first, then add 83 + 52 = 135.

CHAMPION SPEED WAY (Use This!):

Champion Way: Pair the complements 48 + 52 = 100 first, then 100 + 35 = 135!

Archetype 3 — Balanced Equation Shortcut: Solve 76 + 54 = 90 + [?] — Stack Keys: `7` `6` `ENTER` `5` `4` `+` `9` `0` `-`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Add 76 + 54 = 130, then subtract 130 - 90 = 40.

CHAMPION SPEED WAY (Use This!):

Champion Way: Add 76 + 54 = 130 on the stack, then press [90] [-] → 40!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. 9 + [?] = 17 _____

Shortcut Cue: 17 - 9 (or 1 + 7)

W2. 36 + [?] = 60 _____

Shortcut Cue: Count up: +4 to 40, +20 to 60

W3. 82 - 35 = _____

Shortcut Cue: 82 - 30 = 52, then 52 - 5

W4. 64 + 29 + 36 = _____

Shortcut Cue: Pair 64 + 36 = 100 first, then + 29

W5. 200 - 68 - 32 = _____

Shortcut Cue: Combine 68 + 32 = 100, then 200 - 100

W6. 85 + 45 = 100 + [?] _____

Shortcut Cue: 85 + 45 = 130, minus 100

Warm-Up Self-Check Answers: W1: 8 | W2: 24 | W3: 47 | W4: 129 | W5: 100 | W6: 30

Section 2: Number Bonds & Missing Addends

Part-part-whole composition, decomposition, and missing-part equations. • Foundation (Test 2A) & Core Fluency (Test 2B)

TEST 2A • LEVEL 1: FOUNDATION Goal: 3 min

Combining & Decomposing Parts Within 12

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1.	$4 + 3 =$	_____
2.	$5 + 2 =$	_____
3.	$3 + 5 =$	_____
4.	$6 + 2 =$	_____
5.	$4 + 5 =$	_____
6.	$7 + 2 =$	_____
7.	$6 + 3 =$	_____
8.	$7 - 3 =$	_____
9.	$8 - 5 =$	_____
10.	$9 - 4 =$	_____
11.	$9 - 6 =$	_____
12.	$5 + 6 =$	_____
13.	$7 + 4 =$	_____
14.	$8 + 3 =$	_____
15.	$7 + 5 =$	_____
16.	$8 + 4 =$	_____
17.	$9 + 3 =$	_____
18.	$11 - 5 =$	_____
19.	$12 - 7 =$	_____
20.	$12 - 8 =$	_____

TEST 2B • LEVEL 2: CORE FLUENCY Goal: 4 min

Missing Addends & Subtraction Within 25

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1.	$6 + 8 =$	_____
2.	$7 + 6 =$	_____
3.	$8 + 7 =$	_____
4.	$9 + 6 =$	_____
5.	$8 + 9 =$	_____
6.	$13 - 6 =$	_____
7.	$14 - 8 =$	_____
8.	$15 - 7 =$	_____
9.	$16 - 9 =$	_____
10.	$17 - 8 =$	_____
11.	$5 + [\ ?\ ] = 13$	_____
12.	$7 + [\ ?\ ] = 16$	_____
13.	$9 + [\ ?\ ] = 18$	_____
14.	$[\ ?\ ] + 6 = 15$	_____
15.	$[\ ?\ ] + 8 = 19$	_____
16.	$20 - 13 =$	_____
17.	$22 - 8 =$	_____
18.	$14 + [\ ?\ ] = 21$	_____
19.	$24 - 9 =$	_____
20.	$16 + [\ ?\ ] = 25$	_____

Section 2: Number Bonds & Missing Addends

Part-part-whole composition, decomposition, and missing-part equations. • Multi-Step Speed (Test 2C) & Advanced Challenge (Test 2D)

TEST 2C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Two-Digit Number Bonds & Missing Parts

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	24 + 15 =	_____
2.	32 + 27 =	_____
3.	46 + 18 =	_____
4.	38 + 25 =	_____
5.	54 - 21 =	_____
6.	60 - 24 =	_____
7.	72 - 35 =	_____
8.	81 - 46 =	_____
9.	28 + [?] = 50	_____
10.	34 + [?] = 60	_____
11.	[?] + 42 = 75	_____
12.	[?] + 37 = 80	_____
13.	90 - 56 =	_____
14.	85 - 29 =	_____
15.	47 + 36 =	_____
16.	58 + 37 =	_____
17.	63 + [?] = 100	_____
18.	100 - 48 =	_____
19.	110 - 65 =	_____
20.	74 + [?] = 120	_____

TEST 2D • LEVEL 4: ADVANCED CHALLENGE Goal: 5 min

Three-Part Bonds & Balanced Equations

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	125 + 65 =	_____
2.	240 - 85 =	_____
3.	135 + [?] = 200	_____
4.	168 + [?] = 250	_____
5.	320 - 145 =	_____
6.	48 + 35 + 52 =	_____
7.	64 + 29 + 36 =	_____
8.	150 - 45 - 35 =	_____
9.	200 - 68 - 32 =	_____
10.	85 + 45 = 100 + [?]	_____
11.	76 + 54 = 90 + [?]	_____
12.	140 - 55 = 50 + [?]	_____
13.	360 - 175 =	_____
14.	245 + 155 - 80 =	_____
15.	500 - 185 - 115 =	_____
16.	325 + [?] = 600	_____
17.	450 - [?] = 185	_____
18.	(165 + 85) - 70 =	_____
19.	(340 - 160) + 95 =	_____
20.	1000 - 425 - 175 =	_____

Section 3: Complements & Make-10 Bridging

Complements to 10, 100, and 1,000, bridging across tens, and compensation. • Study the Champion Way & complete W1–W6 before starting Tests 3A–3D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 3: COMPLEMENTS & MAKE-10 BRIDGING

RULE 1	**Rule 1 — All-from-9 and the Last-from-10 (Instant Complements to 100 & 1,000): To find 1000 - 384 from left to right, subtract hundreds and tens from 9 (9-3=6, 9-8=1) and ones from 10 (10-4=6) → 616!**
RULE 2	Rule 2 — Fast Compensation for +9, +19, +99, +199: Add the rounded ten/hundred first, then subtract the overshoot (e.g., 498 + 237 = 500 + 235 = 735).
RULE 3	Rule 3 — Decimal Complements to 10 and 100: Treat 10 - 6.75 as 10.00 - 6.75: whole part is 9 - 6 = 3, decimal part is 100 - 75 = .25 → 3.25.

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — Left-to-Right 1,000 Complement: Evaluate 1000 - 742

Stack Keys: `1` `0` `0` `0` `ENTER` `7` `4` `2` `-`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Write 1000 on paper and borrow across three columns from right to left.

CHAMPION SPEED WAY (Use This!):

Champion Way: Apply 9-9-10 from left to right: (9-7=2), (9-4=5), (10-2=8) → write 258 instantly!

Archetype 2 — Near-Hundred Compensation: Evaluate 498 + 237

Stack Keys: `4` `9` `8` `ENTER` `2` `3` `7` `+`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Add column-by-column with two carries (8+7=15, 9+3+1=13, 4+2+1=7).

CHAMPION SPEED WAY (Use This!):

Champion Way: Borrow 2 from 237 to turn 498 into 500: 500 + 235 = 735!

Archetype 3 — Multi-Term Near-Hundred Sum: Evaluate 98 + 97 + 96

Stack Keys: `3` `0` `0` `ENTER` `9` `-`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Add 98 + 97 = 195, then 195 + 96 = 291.

CHAMPION SPEED WAY (Use This!):

Champion Way: Count down from 3 × 100 = 300 by the deficits (2 + 3 + 4 = 9): 300 - 9 = 291!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. 100 - 64 = _____

Shortcut Cue: 9-from-9 (9-6=3), 10-from-10 (10-4=6)

W2. 57 + 19 = _____

Shortcut Cue: Add 20 (77), subtract 1

W3. 1000 - 384 = _____

Shortcut Cue: 9-3=6, 9-8=1, 10-4=6

W4. 10 - 3.65 = _____

Shortcut Cue: 9 - 3 = 6 and 100 - 65 = .35

W5. 299 + 156 = _____

Shortcut Cue: 300 + 155

W6. 99 + 98 + 97 = _____

Shortcut Cue: 300 - (1 + 2 + 3)

Warm-Up Self-Check Answers: W1: 36 | W2: 76 | W3: 616 | W4: 6.35 | W5: 455 | W6: 294

Section 3: Complements & Make-10 Bridging

Complements to 10, 100, and 1,000, bridging across tens, and compensation. • Foundation (Test 3A) & Core Fluency (Test 3B)

TEST 3A • LEVEL 1: FOUNDATION Goal: 3 min

Complements to 10 & Teen Place Value

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	10 - 7 =	_____
2.	10 - 4 =	_____
3.	10 - 8 =	_____
4.	10 - 3 =	_____
5.	10 - 9 =	_____
6.	10 - 6 =	_____
7.	10 - 2 =	_____
8.	10 - 5 =	_____
9.	10 + 4 =	_____
10.	10 + 7 =	_____
11.	10 + 9 =	_____
12.	8 + 2 + 3 =	_____
13.	9 + 1 + 5 =	_____
14.	7 + 3 + 4 =	_____
15.	6 + 4 + 2 =	_____
16.	8 + 5 =	_____
17.	9 + 6 =	_____
18.	7 + 5 =	_____
19.	8 + 6 =	_____
20.	9 + 7 =	_____

TEST 3B • LEVEL 2: CORE FLUENCY Goal: 4 min

Bridging Across 10, 20, and 50

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	9 + 8 =	_____
2.	8 + 7 =	_____
3.	7 + 6 =	_____
4.	9 + 5 =	_____
5.	20 - 7 =	_____
6.	20 - 4 =	_____
7.	20 - 12 =	_____
8.	18 + 5 =	_____
9.	19 + 7 =	_____
10.	17 + 6 =	_____
11.	28 + 6 =	_____
12.	29 + 8 =	_____
13.	37 + 5 =	_____
14.	30 - 6 =	_____
15.	40 - 8 =	_____
16.	50 - 14 =	_____
17.	50 - 27 =	_____
18.	48 + 7 =	_____
19.	59 + 8 =	_____
20.	70 - 23 =	_____

Section 3: Complements & Make-10 Bridging

Complements to 10, 100, and 1,000, bridging across tens, and compensation. • Multi-Step Speed (Test 3C) & Advanced Challenge (Test 3D)

TEST 3C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Complements to 100 & Fast Compensation (+9, +19, +29)

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1.	100 - 30 =	_____
2.	100 - 45 =	_____
3.	100 - 62 =	_____
4.	100 - 28 =	_____
5.	100 - 74 =	_____
6.	100 - 19 =	_____
7.	100 - 83 =	_____
8.	100 - 57 =	_____
9.	36 + 9 =	_____
10.	58 + 9 =	_____
11.	47 + 19 =	_____
12.	65 + 19 =	_____
13.	38 + 29 =	_____
14.	54 + 39 =	_____
15.	73 - 9 =	_____
16.	82 - 19 =	_____
17.	95 - 29 =	_____
18.	67 + 33 + 25 =	_____
19.	48 + 52 + 39 =	_____
20.	200 - 64 =	_____

TEST 3D • LEVEL 4: ADVANCED CHALLENGE Goal: 5 min

Complements to 1,000, Decimals & Near-Hundred Sums

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1.	1000 - 400 =	_____
2.	1000 - 650 =	_____
3.	1000 - 275 =	_____
4.	1000 - 384 =	_____
5.	1000 - 742 =	_____
6.	1000 - 519 =	_____
7.	1000 - 863 =	_____
8.	1000 - 97 =	_____
9.	10 - 3.4 =	_____
10.	10 - 6.75 =	_____
11.	100 - 42.5 =	_____
12.	99 + 47 =	_____
13.	98 + 65 =	_____
14.	199 + 84 =	_____
15.	299 + 156 =	_____
16.	498 + 237 =	_____
17.	99 + 99 + 99 =	_____
18.	98 + 97 + 96 =	_____
19.	198 + 297 + 105 =	_____
20.	5000 - 1645 =	_____

Section 4: Doubles, Equal Groups & Squares

Doubles, near-doubles, equal groups, halving, and mental squaring. • Study the Champion Way & complete W1–W6 before starting Tests 4A–4D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 4: DOUBLES, EQUAL GROUPS & SQUARES

RULE 1	**Rule 1 — Stack Doubling & Squaring Reflex: Pressing [ENTER] duplicates Line X into Line Y, so [N] [ENTER] [+] doubles any number and [N] [ENTER] [×] (or [x^2]) squares it in 2–3 keys!**
RULE 2	Rule 2 — Squaring Numbers Ending in 5 ($n5^2$): Multiply the tens digit n by (n + 1) and append 25! Example: $35^2 \rightarrow 3 \times 4 = 12$, append 25 → 1225.
RULE 3	Rule 3 — Near-Square Product $(a - 1)(a + 1) = a^2 - 1$: Whenever two factors straddle a round number (like 19 × 21 or 29 × 31), square the middle number and subtract 1 ($20^2 - 1 = 399$; $30^2 - 1 = 899$).

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — Doubling & Halving Chain: Evaluate 4 × 35 and 130 ÷ 2

Stack Keys: 3 5 ENTER + ENTER +

SLOW / AMATEUR WAY (Avoid):

Slow Way: Multiply 35 × 4 with paper carries; do long division for 130 ÷ 2.

CHAMPION SPEED WAY (Use This!):

Champion Way: Double twice for ×4 (35 → 70 → 140); halve 100 + 30 → 50 + 15 = 65!

Archetype 2 — Ends-in-5 Mental Square: Evaluate 25^2 and 35^2

Stack Keys: 3 5 ENTER ×

SLOW / AMATEUR WAY (Avoid):

Slow Way: Multiply 35 × 35 by hand (4 partial products).

CHAMPION SPEED WAY (Use This!):

Champion Way: For 25^2, $2 \times 3 = 6 \rightarrow 625$. For 35^2, $3 \times 4 = 12 \rightarrow 1225$!

Archetype 3 — Difference of Squares Shortcut: Evaluate $25^2 - 15^2$

Stack Keys: 2 5 ENTER 1 5 + 1 0 ×

SLOW / AMATEUR WAY (Avoid):

Slow Way: Compute 625 and 225 separately, then subtract.

CHAMPION SPEED WAY (Use This!):

Champion Way: Use $(a + b)(a - b) = (25 + 15)(25 - 15) = 40 \times 10 = 400$ in 1 second!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. $2 \times 46 =$ _____

Shortcut Cue: Double 40 (80) + double 6 (12)

W2. $4 \times 18 =$ _____

Shortcut Cue: Double 18 → 36, double 36 → 72

W3. $15^2 =$ _____

Shortcut Cue: Ends-in-5 rule: 1 × 2 = 2, append 25

W4. $25^2 =$ _____

Shortcut Cue: Ends-in-5 rule: 2 × 3 = 6, append 25

W5. $19 \times 21 =$ _____

Shortcut Cue: $20^2 - 1 = 400 - 1$

W6. $15^2 - 5^2 =$ _____

Shortcut Cue: (15 + 5)(15 - 5) = 20 × 10

Warm-Up Self-Check Answers: W1: 92 | W2: 72 | W3: 225 | W4: 625 | W5: 399 | W6: 200

Section 4: Doubles, Equal Groups & Squares

Doubles, near-doubles, equal groups, halving, and mental squaring. • Foundation (Test 4A) & Core Fluency (Test 4B)

TEST 4A • LEVEL 1: FOUNDATION Goal: 3 min

Doubles and Near-Doubles Within 25

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	$3 + 3 =$	
2.	$4 + 4 =$	
3.	$5 + 5 =$	
4.	$6 + 6 =$	
5.	$7 + 7 =$	
6.	$8 + 8 =$	
7.	$9 + 9 =$	
8.	$10 + 10 =$	
9.	$4 + 5 =$	
10.	$5 + 6 =$	
11.	$6 + 7 =$	
12.	$7 + 8 =$	
13.	$8 + 9 =$	
14.	$9 + 10 =$	
15.	$11 + 11 =$	
16.	$12 + 12 =$	
17.	$10 + 11 =$	
18.	$11 + 12 =$	
19.	$2 \times 6 =$	
20.	$2 \times 9 =$	

TEST 4B • LEVEL 2: CORE FLUENCY Goal: 4 min

Two-Digit Doubles, Halves & Equal Groups

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	$13 + 13 =$	
2.	$14 + 14 =$	
3.	$15 + 15 =$	
4.	$16 + 16 =$	
5.	$18 + 18 =$	
6.	$25 + 25 =$	
7.	$35 + 35 =$	
8.	$45 + 45 =$	
9.	$15 + 16 =$	
10.	$24 + 25 =$	
11.	$3 \times 4 =$	
12.	$4 \times 5 =$	
13.	$3 \times 6 =$	
14.	$5 \times 6 =$	
15.	$4 \times 8 =$	
16.	$6 \times 7 =$	
17.	$24 \div 2 =$	
18.	$36 \div 2 =$	
19.	$50 \div 2 =$	
20.	$48 \div 2 =$	

Section 4: Doubles, Equal Groups & Squares

Doubles, near-doubles, equal groups, halving, and mental squaring. • Multi-Step Speed (Test 4C) & Advanced Challenge (Test 4D)

TEST 4C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Doubling Chains (×2, ×4, ×8) & Two-Digit Multiplication

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1.	$2 \times 27 =$	
2.	$2 \times 38 =$	
3.	$2 \times 46 =$	
4.	$2 \times 65 =$	
5.	$2 \times 78 =$	
6.	$4 \times 15 =$	
7.	$4 \times 18 =$	
8.	$4 \times 23 =$	
9.	$4 \times 25 =$	
10.	$4 \times 35 =$	
11.	$8 \times 12 =$	
12.	$8 \times 15 =$	
13.	$8 \times 25 =$	
14.	$74 \div 2 =$	
15.	$96 \div 2 =$	
16.	$130 \div 2 =$	
17.	$84 \div 4 =$	
18.	$120 \div 4 =$	
19.	$2 \times 34 + 10 =$	
20.	$4 \times 16 - 14 =$	

TEST 4D • LEVEL 4: ADVANCED CHALLENGE Goal: 5 min

Mental Squares, Near-Squares (a-1)(a+1) & Powers

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1.	$11^2 =$	
2.	$12^2 =$	
3.	$13^2 =$	
4.	$14^2 =$	
5.	$15^2 =$	
6.	$16^2 =$	
7.	$20^2 =$	
8.	$25^2 =$	
9.	$30^2 =$	
10.	$35^2 =$	
11.	$9 \times 11 =$	
12.	$11 \times 13 =$	
13.	$14 \times 16 =$	
14.	$19 \times 21 =$	
15.	$24 \times 26 =$	
16.	$29 \times 31 =$	
17.	$2 \times 15^2 =$	
18.	$10^2 + 12^2 =$	
19.	$15^2 - 5^2 =$	
20.	$25^2 - 15^2 =$	

Cumulative Review C1: Number Sense (Sections 1–4)

Integrated review across step counting, number bonds, complements, and doubles. • Study the Champion Way & complete W1–W6 before starting Tests C1A–C1D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — CUMULATIVE REVIEW C1: NUMBER SENSE (SECTIONS 1–4)

RULE 1	**Rule 1 — Scan Before Computing: Before pressing a key or writing a digit, check if the problem is a 10/100 complement, a Gauss series, an ends-in-5 square, or an (a - 1)(a + 1) near-square.**
RULE 2	Rule 2 — Never Write Intermediate Steps on Single-Register Arithmetic: Let Register Y hold your running total while you enter the next step on Register X.
RULE 3	Rule 3 — Use Factor Splitting for Mental Multiplication: Treat ×4 as double-double, ×8 as double-double-double, and ÷4 as half-half.

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — Combined Series & Complement: Evaluate 99 + 98 + 97 + 96 Stack Keys: [4] [0] [0] [ENTER] [1] [0] [-]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Add four 2-digit numbers one by one (99+98=197, +97=294, +96=390).

CHAMPION SPEED WAY (Use This!):

Champion Way: 4 × 100 minus the deficits (1 + 2 + 3 + 4 = 10): 400 - 10 = 390!

Archetype 2 — Difference of Squares Mental Sprint: Evaluate $18^2 - 8^2$ Stack Keys: [1] [8] [ENTER] [8] [+] [1] [0] [×]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Calculate $18^2 = 324$ and $8^2 = 64$, then subtract.

CHAMPION SPEED WAY (Use This!):

Champion Way: (18 + 8) × (18 - 8) = 26 × 10 = 260 immediately!

Archetype 3 — Multi-Step Stack Chain: Evaluate 2 × 20^2 - 150 Stack Keys: [2] [0] [ENTER] [×] [2] [×] [1] [5] [0] [-]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Compute 20^2, write 400, multiply by 2, write 800, subtract 150.

CHAMPION SPEED WAY (Use This!):

Champion Way: Mental 2 × 400 - 150 = 650, or key [20] [ENTER] [×] [2] [×] [150] [-]!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. 1000 - 465 = _____

Shortcut Cue: Left-to-right 9-9-10: 5, 3, 5

W2. 398 + 165 = _____

Shortcut Cue: 400 + 163

W3. 29 × 31 = _____

Shortcut Cue: $30^2 - 1 = 900 - 1$

W4. 420 ÷ 4 = _____

Shortcut Cue: Halve twice: 420 → 210 → 105

W5. 1 + 3 + 5 + 7 + 9 + 11 + 13 = _____

Shortcut Cue: 7 odd terms → 7^2

W6. 45 + 55 + 65 + 75 = _____

Shortcut Cue: Pair (45+75) + (55+65) = 120 + 120

Warm-Up Self-Check Answers: W1: 535 | W2: 563 | W3: 899 | W4: 105 | W5: 49 | W6: 240

Cumulative Review C1: Number Sense (Sections 1–4)

Integrated review across step counting, number bonds, complements, and doubles. • Foundation (Test C1A) & Core Fluency (Test C1B)

TEST C1A • LEVEL 1: FOUNDATION Goal: 4 min

Core Addition, Subtraction, Complements to 10 & Doubles

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	$6 + 1 + 1 =$	_____
2.	$12 - 2 =$	_____
3.	$5 + 4 =$	_____
4.	$11 - 6 =$	_____
5.	$10 - 8 =$	_____
6.	$10 - 3 =$	_____
7.	$10 + 6 =$	_____
8.	$7 + 7 =$	_____
9.	$8 + 8 =$	_____
10.	$6 + 7 =$	_____
11.	$8 + 5 =$	_____
12.	$9 + 6 =$	_____
13.	$14 - 7 =$	_____
14.	$16 - 8 =$	_____
15.	$5 + 5 + 5 =$	_____
16.	$9 + 9 =$	_____
17.	$7 + 8 =$	_____
18.	$15 - 6 =$	_____
19.	$10 + 10 =$	_____
20.	$20 - 5 =$	_____

TEST C1B • LEVEL 2: CORE FLUENCY Goal: 4 min

Two-Digit Fluency, Missing Addends & Equal Groups

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	$18 + 2 + 7 =$	_____
2.	$35 - 5 - 10 =$	_____
3.	$8 + [\ ?\ ] = 17$	_____
4.	$14 + [\ ?\ ] = 22$	_____
5.	$30 - 14 =$	_____
6.	$50 - 18 =$	_____
7.	$16 + 16 =$	_____
8.	$24 + 24 =$	_____
9.	$17 + 18 =$	_____
10.	$3 \times 7 =$	_____
11.	$4 \times 6 =$	_____
12.	$5 \times 8 =$	_____
13.	$32 \div 2 =$	_____
14.	$46 \div 2 =$	_____
15.	$29 + 7 =$	_____
16.	$48 + 9 =$	_____
17.	$60 - 25 =$	_____
18.	$25 + 25 + 15 =$	_____
19.	$4 \times 9 =$	_____
20.	$100 - 45 =$	_____

Cumulative Review C1: Number Sense (Sections 1–4)

Integrated review across step counting, number bonds, complements, and doubles. • Multi-Step Speed (Test C1C) & Advanced Challenge (Test C1D)

TEST C1C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Complements to 100, Doubling Chains & Multi-Step Bonds

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1.	$100 - 37 =$	
2.	$100 - 68 =$	
3.	$46 + 38 =$	
4.	$83 - 47 =$	
5.	$57 + 19 =$	
6.	$84 - 29 =$	
7.	$2 \times 48 =$	
8.	$4 \times 19 =$	
9.	$8 \times 15 =$	
10.	$110 \div 2 =$	
11.	$92 \div 4 =$	
12.	$64 + [\ ?\ ] = 120$	
13.	$150 - 85 =$	
14.	$35 + 45 + 20 =$	
15.	$12^2 =$	
16.	$15^2 =$	
17.	$2 \times 36 + 18 =$	
18.	$100 - 24 - 36 =$	
19.	$250 - 125 + 40 =$	
20.	$4 \times 25 - 33 =$	

TEST C1D • LEVEL 4: ADVANCED CHALLENGE Goal: 5 min

Series Sums, Near-Squares & Mental Compensation

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1.	$1 + 2 + 3 + 4 + 5 + 6 =$	
2.	$2 + 4 + 6 + 8 + 10 + 12 =$	
3.	$1000 - 465 =$	
4.	$1000 - 782 =$	
5.	$199 + 247 =$	
6.	$398 + 165 =$	
7.	$14^2 =$	
8.	$25^2 =$	
9.	$19 \times 21 =$	
10.	$29 \times 31 =$	
11.	$10 - 4.35 =$	
12.	$100 - 63.8 =$	
13.	$99 + 98 + 97 + 96 =$	
14.	$18^2 - 8^2 =$	
15.	$45 + 55 + 65 + 75 =$	
16.	$8 \times 35 =$	
17.	$420 \div 4 =$	
18.	$350 + [\ ?\ ] = 825$	
19.	$2 \times 20^2 - 150 =$	
20.	$1 + 3 + 5 + 7 + 9 + 11 + 13 =$	

Section 5: Multi-Step Operations & Grouping

Multiplication, division, inside-out parentheses, and two-branch expressions. • Study the Champion Way & complete W1–W6 before starting Tests 5A–5D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 5: MULTI-STEP OPERATIONS & GROUPING

RULE 1	**Rule 1 — Start at the Deepest Inner Parentheses: Never type left-to-right when an inner group is on the right! Evaluate the deepest inner group first so it sits on the stack.**
RULE 2	Rule 2 — Automatic Stack Lift Saves [ENTER] Keys: After any operation (+, -, ×, ÷), typing the next number automatically lifts the result to Shelf Y—never press [ENTER] after an operator!
RULE 3	Rule 3 — Two-Branch Expression Pipeline (a ± b) × (c ± d): Compute (a ± b) first, then immediately type c [ENTER] d [±] (which lifts the first branch to Y), and press [×] or [÷].

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — Right-Hand Inner Group: Evaluate 100 - (16 × 5 - 12)

Stack Keys: `1` `6` `ENTER` `5` `×` `1` `2` `-` `1` `0` `0` `⇅` `-`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Type 100 [ENTER] 16 [ENTER] 5 [×] [12] [-] [-] (extra stack depth & mental tracking).

CHAMPION SPEED WAY (Use This!):

Champion Way: Compute inner 16 × 5 - 12 = 68 first, then subtract from 100: 100 - 68 = 32!

Archetype 2 — Two Parallel Branches: Evaluate (50 - 14) ÷ (3 + 6)

Stack Keys: `5` `0` `ENTER` `1` `4` `-` `3` `ENTER` `6` `+` `÷`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Compute 50 - 14 = 36, write 36 on paper, compute 3 + 6 = 9, re-type 36 ÷ 9.

CHAMPION SPEED WAY (Use This!):

Champion Way: Key [50] [ENTER] [14] [-] (36 in X), then [3] [ENTER] [6] [+] (36 lifts to Y, 9 in X), then [÷] → 4!

Archetype 3 — Common Factor Distribution: Evaluate (28 × 9) - (18 × 9)

Stack Keys: `2` `8` `ENTER` `1` `8` `-` `9` `×`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Multiply 28 × 9 = 252 and 18 × 9 = 162, then subtract.

CHAMPION SPEED WAY (Use This!):

Champion Way: Factor out ×9 mentally: (28 - 18) × 9 = 10 × 9 = 90 in 1 second!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. $(18 - 6) \times 3 =$ _____

Shortcut Cue: [18] [ENTER] [6] [-] [3] [×] (no extra ENTER!)

W2. $45 \div (12 - 3) =$ _____

Shortcut Cue: Inner 12 - 3 = 9 first, then 45 ÷ 9

W3. $(15 \times 4) + (18 \div 2) =$ _____

Shortcut Cue: 60 + 9 (or two-branch stack)

W4. $((20 - 8) \times 4) \div 6 =$ _____

Shortcut Cue: 12 × 4 = 48, then ÷ 6

W5. $(17 \times 6) - (13 \times 6) =$ _____

Shortcut Cue: Factor out 6: (17 - 13) × 6

W6. $(45 \times 12) \div 18 =$ _____

Shortcut Cue: Cancel 9: (5 × 12) ÷ 2

Warm-Up Self-Check Answers: W1: 36 | W2: 5 | W3: 69 | W4: 8 | W5: 24 | W6: 30

Section 5: Multi-Step Operations & Grouping

Multiplication, division, inside-out parentheses, and two-branch expressions. • Foundation (Test 5A) & Core Fluency (Test 5B)

TEST 5A • LEVEL 1: FOUNDATION Goal: 4 min

Core Multiplication, Division & Two-Step Expressions

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	$6 \times 4 =$	
2.	$7 \times 5 =$	
3.	$8 \times 6 =$	
4.	$9 \times 7 =$	
5.	$24 \div 3 =$	
6.	$36 \div 4 =$	
7.	$45 \div 5 =$	
8.	$56 \div 8 =$	
9.	$(10 - 4) \times 3 =$	
10.	$(12 + 8) \div 4 =$	
11.	$(15 - 6) \times 2 =$	
12.	$(18 + 6) \div 3 =$	
13.	$5 \times 4 + 7 =$	
14.	$6 \times 5 - 8 =$	
15.	$20 - 3 \times 4 =$	
16.	$30 - 4 \times 5 =$	
17.	$14 + 18 \div 3 =$	
18.	$25 - 20 \div 4 =$	
19.	$(9 + 6) \times 2 =$	
20.	$40 \div (3 + 5) =$	

TEST 5B • LEVEL 2: CORE FLUENCY Goal: 4 min

Order of Operations & Inner Parentheses

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	$(18 - 6) \times 3 =$	
2.	$(24 + 16) \div 8 =$	
3.	$45 \div (12 - 3) =$	
4.	$60 \div (7 + 5) =$	
5.	$12 + 6 \times 4 =$	
6.	$50 - 7 \times 6 =$	
7.	$8 \times 7 - 16 =$	
8.	$9 \times 8 + 18 =$	
9.	$(32 - 12) \times 4 =$	
10.	$(48 + 12) \div 6 =$	
11.	$72 \div (4 + 5) =$	
12.	$100 - 15 \times 4 =$	
13.	$24 + 36 \div 6 =$	
14.	$64 \div 8 + 17 =$	
15.	$(14 + 7) \div 3 \times 5 =$	
16.	$(25 - 10) \times 4 \div 6 =$	
17.	$90 \div (15 - 6) \times 4 =$	
18.	$120 - (8 \times 9) =$	
19.	$(16 + 24) \div (12 - 4) =$	
20.	$(15 - 3) \times (2 + 4) =$	

Section 5: Multi-Step Operations & Grouping

Multiplication, division, inside-out parentheses, and two-branch expressions. • Multi-Step Speed (Test 5C) & Advanced Challenge (Test 5D)

TEST 5C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Two-Branch Expressions (a ± b) × (c ± d) & Quotients

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1.	(15 × 4) + (18 ÷ 2) =	
2.	(12 × 5) - (8 × 4) =	
3.	(50 - 14) ÷ (3 + 6) =	
4.	(42 + 18) ÷ (11 - 5) =	
5.	(8 + 4) × (9 - 3) =	
6.	(14 - 5) × (6 + 2) =	
7.	(20 × 3) + (15 × 2) =	
8.	(16 × 4) - (12 × 3) =	
9.	((20 - 8) × 4) ÷ 6 =	
10.	((18 + 6) ÷ 3) × 7 =	
11.	(72 ÷ (4 + 5)) × 3 =	
12.	100 - (16 × 5 - 12) =	
13.	(84 ÷ 7) + (45 ÷ 9) =	
14.	(65 - 25) × (18 ÷ 6) =	
15.	((8 + 4) × (9 - 3)) ÷ 12 =	
16.	(96 ÷ 8) × (20 - 15) =	
17.	150 - (12 × 6 + 18) =	
18.	(35 + 25) ÷ (4 × 3) =	
19.	(18 × 5) ÷ (14 - 5) =	
20.	(24 - 9) × (16 - 8) =	

TEST 5D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Multi-Digit Chained Operations & Deep Nesting

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1.	(25 × 8) - (15 × 6) =	
2.	(144 ÷ 12) × (36 ÷ 4) =	
3.	(125 + 75) ÷ (32 - 7) =	
4.	((45 - 15) × 6) ÷ (12 + 6) =	
5.	250 - (18 × 7 + 24) =	
6.	(36 × 5) ÷ (27 - 9) =	
7.	((64 ÷ 4) + 9) × 4 =	
8.	(180 ÷ (12 + 3)) × 8 =	
9.	(22 + 18) × (35 - 20) =	
10.	(160 - 40) ÷ (6 × 4) =	
11.	((32 - 14) × 5) ÷ 6 =	
12.	80 - (15 × 4) + 12 =	
13.	(240 ÷ 15) + (120 ÷ 8) =	
14.	(17 × 6) - (13 × 6) =	
15.	(28 × 9) - (18 × 9) =	
16.	(45 × 12) ÷ 18 =	
17.	((100 - 28) ÷ 9) × 15 =	
18.	(360 ÷ (18 - 6)) - 14 =	
19.	((15 + 5) × (8 - 3)) ÷ 10 =	
20.	(500 - 140) ÷ (9 × 4) =	

Section 6: Fractions, Decimals & Reciprocals

Fraction equivalence, unlike denominators, mixed numbers, decimals, and reciprocals. • Study the Champion Way & complete W1–W6 before starting Tests 6A–6D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 6: FRACTIONS, DECIMALS & RECIPROCALS

RULE 1	**Rule 1 — Harmonic & Reciprocal Sums via [1/x]: To evaluate 1 ÷ (1/a + 1/b) on the calculator, never find a common denominator by hand—key [a] [1/x] [b] [1/x] [+] [1/x] in 6 keys!**
RULE 2	Rule 2 — Distribute Whole Multipliers to Clear Denominators: For 24 × (5/6 - 3/8), multiply 24 into each fraction directly: (24 × 5/6) - (24 × 3/8) = 20 - 9 = 11!
RULE 3	Rule 3 — Memorize Eighths & Thirds as Instant Decimals: 1/8 = 0.125, 3/8 = 0.375, 5/8 = 0.625, 7/8 = 0.875; and use Two-Dot entry ([2][.][1][.][2] = 2 1/2) on the SC-6.

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — Harmonic Parallel Sum: Evaluate 1 ÷ (1/4 + 1/6) Stack Keys: [4] [x^{-1}] [6] [x^{-1}] [+] [x^{-1}]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Convert to twelfths (3/12 + 2/12 = 5/12), invert to 12/5, divide 12 ÷ 5 = 2.4.

CHAMPION SPEED WAY (Use This!):

Champion Way: Use the 6-key Reciprocal Pipeline: [4] [1/x] [6] [1/x] [+] [1/x] → 2.4!

Archetype 2 — Clearing Denominators by Distribution: Evaluate 24 × (5/6 - 3/8) Stack Keys: [2] [0] [ENTER] [9] [-]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Subtract 20/24 - 9/24 = 11/24, then multiply by 24.

CHAMPION SPEED WAY (Use This!):

Champion Way: Distribute 24 immediately: 24(5/6) = 20 and 24(3/8) = 9, so 20 - 9 = 11!

Archetype 3 — Telescoping Product Cancellation: Evaluate 1/2 × 2/3 × 3/4 × 4/5 Stack Keys: [1] [ENTER] [5] [÷]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Multiply numerators (1×2×3×4 = 24) and denominators (2×3×4×5 = 120) and reduce.

CHAMPION SPEED WAY (Use This!):

Champion Way: Cancel interior 2, 3, 4 diagonally—only first numerator (1) and last denominator (5) survive → 1/5!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. $3/4 \div 1/2$ (decimal) = _____

Shortcut Cue: Multiply by reciprocal: 3/4 × 2 = 1.5

W2. $3/4 + (1/2 \times 1/4)$ (decimal) = _____

Shortcut Cue: 0.75 + 0.125

W3. $1 \div (1/3 + 1/6) =$ _____

Shortcut Cue: 1/3 + 1/6 = 1/2, reciprocal is 2

W4. $(1/3 + 1/4) \times 24 =$ _____

Shortcut Cue: Distribute 24: 8 + 6

W5. $2/3 \times 3/4 \times 4/5 =$ _____

Shortcut Cue: Telescoping cancel 3 and 4 → 2/5

W6. $36 \times (7/9 - 1/4) =$ _____

Shortcut Cue: Distribute 36: 28 - 9

Warm-Up Self-Check Answers: W1: 1.5 | W2: 0.875 | W3: 2 | W4: 14 | W5: 2/5 | W6: 19

Section 6: Fractions, Decimals & Reciprocals

Fraction equivalence, unlike denominators, mixed numbers, decimals, and reciprocals. • Foundation (Test 6A) & Core Fluency (Test 6B)

TEST 6A • LEVEL 1: FOUNDATION Goal: 4 min

Like Fractions, Halves/Fourths & Exact Decimals

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	1/4 + 2/4 =	_____
2.	3/5 - 1/5 =	_____
3.	3/8 + 2/8 =	_____
4.	7/10 - 4/10 =	_____
5.	1/2 + 1/4 =	_____
6.	3/4 - 1/2 =	_____
7.	1/2 + 1/2 =	_____
8.	1 - 1/4 =	_____
9.	1 - 3/8 =	_____
10.	1/2 as a decimal =	_____
11.	1/4 as a decimal =	_____
12.	3/4 as a decimal =	_____
13.	1/5 as a decimal =	_____
14.	3/5 as a decimal =	_____
15.	1/10 + 3/10 (simplified) =	_____
16.	2/6 + 1/6 (simplified) =	_____
17.	1/2 of 16 =	_____
18.	1/4 of 20 =	_____
19.	3/4 of 12 =	_____
20.	2/3 of 15 =	_____

TEST 6B • LEVEL 2: CORE FLUENCY Goal: 5 min

Unlike Denominators, Multiplication & Reciprocals

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	1/3 + 1/6 =	_____
2.	2/3 - 1/6 =	_____
3.	3/8 + 1/4 =	_____
4.	5/6 - 1/3 =	_____
5.	1/2 + 1/3 =	_____
6.	1/2 - 1/3 =	_____
7.	1/3 + 1/4 =	_____
8.	3/4 - 1/3 =	_____
9.	1/2 × 3/4 =	_____
10.	2/3 × 3/4 =	_____
11.	3/5 × 5/6 =	_____
12.	4/5 × 15 =	_____
13.	Reciprocal of 3/4 =	_____
14.	Reciprocal of 4/5 (decimal) =	_____
15.	Reciprocal of 2/5 (decimal) =	_____
16.	1/2 ÷ 1/4 =	_____
17.	3/4 ÷ 1/4 =	_____
18.	3/4 ÷ 1/2 (decimal) =	_____
19.	2/3 ÷ 2/3 =	_____
20.	6 ÷ 1/2 =	_____

Section 6: Fractions, Decimals & Reciprocals

Fraction equivalence, unlike denominators, mixed numbers, decimals, and reciprocals. • Multi-Step Speed (Test 6C) & Advanced Challenge (Test 6D)

TEST 6C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Mixed Numbers, Decimal Equivalents & Fraction Chains

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. **2 1/2 + 1 1/4 (decimal) =** _____
2. **3 3/4 - 1 1/2 (decimal) =** _____
3. **5 - 2 3/4 (decimal) =** _____
4. **3 1/2 - 1 3/4 (decimal) =** _____
5. **1 1/2 × 4 =** _____
6. **2 1/2 × 6 =** _____
7. **1 1/4 × 8 =** _____
8. **3 1/3 × 3 =** _____
9. **(1/3 + 1/6) × 12 =** _____
10. **(1/2 + 1/4) × 8 =** _____
11. **(3/4 - 1/8) × 16 =** _____
12. **3/4 + (1/2 × 1/4) (decimal) =** _____
13. **1/8 as a decimal =** _____
14. **3/8 as a decimal =** _____
15. **5/8 as a decimal =** _____
16. **7/8 as a decimal =** _____
17. **4 1/2 ÷ 1 1/2 =** _____
18. **2 1/4 ÷ 3/4 =** _____
19. **0.25 + 0.5 + 0.75 =** _____
20. **1.25 × 8 =** _____

TEST 6D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Harmonic Means, Complex Fractions & Multi-Step Rationals

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. **1 ÷ (1/2 + 1/2) =** _____
2. **1 ÷ (1/3 + 1/6) =** _____
3. **1 ÷ (1/4 + 1/4) =** _____
4. **1 ÷ (1/4 + 1/6) (decimal) =** _____
5. **1 ÷ (1/3 + 1/3 + 1/3) =** _____
6. **(1/2 + 1/3) × 12 =** _____
7. **(1/3 + 1/4) × 24 =** _____
8. **(3/4 - 2/3) × 12 =** _____
9. **(2/3 + 3/4) × 12 =** _____
10. **(1/2 + 1/3 + 1/6) × 18 =** _____
11. **2/3 × 3/4 × 4/5 =** _____
12. **1/2 × 2/3 × 3/4 × 4/5 =** _____
13. **(3/4) ÷ (3/8) =** _____
14. **(5/6) ÷ (5/12) =** _____
15. **(2 1/2 + 1 1/2) ÷ (1/4) =** _____
16. **1 + 1/(1 + 1/2) =** _____
17. **1 + 1/(1 + 1/3) =** _____
18. **(3 1/3 ÷ 1 1/3) (decimal) =** _____
19. **24 × (5/6 - 3/8) =** _____
20. **36 × (7/9 - 1/4) =** _____

Section 7: Primes, GCF/LCM & Nested Expressions

Prime factorization, GCF, LCM, powers, and multi-branch expression trees. • Study the Champion Way & complete W1–W6 before starting Tests 7A–7D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 7: PRIMES, GCF/LCM & NESTED EXPRESSIONS

RULE 1	**Rule 1 — Product Identity GCF(a, b) × LCM(a, b) = a × b: Never compute GCF and LCM separately when asked for their product—just multiply a × b directly!**
RULE 2	Rule 2 — Divisor Count Formula from Prime Exponents: If $N = p^a \times q^b$, the number of positive factors is $(a + 1)(b + 1)$. Example: $24 = 2^3 \times 3^1 \rightarrow (3 + 1)(1 + 1) = 8$ factors.
RULE 3	Rule 3 — Difference of Squares Quotients $(a^2 - b^2) \div (a \pm b)$: Simplify algebraically before touching the calculator: $(a^2 - b^2) \div (a - b) = a + b$, and $(a^2 - b^2) \div (a + b) = a - b$!

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — GCF × LCM Identity: Evaluate GCF(15, 20) × LCM(15, 20) Stack Keys: `1` `5` `ENTER` `2` `0` `×`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Find GCF = 5 and LCM = 60, then multiply 5 × 60 = 300.

CHAMPION SPEED WAY (Use This!):

Champion Way: Use GCF(a, b) × LCM(a, b) = a × b → 15 × 20 = 300 in 2 seconds!

Archetype 2 — Counting Positive Divisors: How many positive factors does 36 have? Stack Keys: `3` `ENTER` `3` `×`

SLOW / AMATEUR WAY (Avoid):

Slow Way: List 1, 2, 3, 4, 6, 9, 12, 18, 36 by trial division (easy to miss one).

CHAMPION SPEED WAY (Use This!):

Champion Way: Prime factor 36 = $2^2 \times 3^2$, then add 1 to each exponent: (2 + 1)(2 + 1) = 9!

Archetype 3 — Algebraic Quotient Shortcut: Evaluate $(25^2 - 15^2) \div (25 - 15)$ Stack Keys: `2` `5` `ENTER` `1` `5` `+`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Compute 625 - 225 = 400, then divide by 10 = 40.

CHAMPION SPEED WAY (Use This!):

Champion Way: Cancel (25 - 15) immediately to leave 25 + 15 = 40!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. $GCF(12, 18) \times LCM(12, 18) =$ _____

Shortcut Cue: Use $a \times b = 12 \times 18$

W2. Number of positive factors of 24 = _____

Shortcut Cue: $24 = 2^3 \times 3^1 \rightarrow (3+1)(1+1)$

W3. $(15^2 - 5^2) \div (15 + 5) =$ _____

Shortcut Cue: Simplifies to 15 - 5

W4. $(30 - (4 \times 5)) \times (12 \div (2 + 4)) =$ _____

Shortcut Cue: $(30 - 20) \times (12 \div 6) = 10 \times 2$

W5. $(2^5 - 2^3) \div (3^2 - 3) =$ _____

Shortcut Cue: $(32 - 8) \div (9 - 3) = 24 \div 6$

W6. $(12^2 + 5^2) \div 13 =$ _____

Shortcut Cue: 169 ÷ 13 (since 5-12-13 → $13^2 \div 13$)

Warm-Up Self-Check Answers: W1: 216 | W2: 8 | W3: 10 | W4: 20 | W5: 4 | W6: 13

Section 7: Primes, GCF/LCM & Nested Expressions

Prime factorization, GCF, LCM, powers, and multi-branch expression trees. • Foundation (Test 7A) & Core Fluency (Test 7B)

TEST 7A • LEVEL 1: FOUNDATION Goal: 4 min

Prime Powers, Squares, Cubes & Factor Pairs

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $2 \times 2 \times 3 =$ _____
2. $2 \times 3 \times 3 =$ _____
3. $2 \times 2 \times 5 =$ _____
4. $2 \times 2 \times 2 \times 3 =$ _____
5. $2 \times 3 \times 5 =$ _____
6. $2 \times 2 \times 3 \times 3 =$ _____
7. $2^3 =$ _____
8. $2^4 =$ _____
9. $3^3 =$ _____
10. $4^3 =$ _____
11. $5^3 =$ _____
12. Smallest prime number = _____
13. 7th prime (2,3,5,7,11,13,?) = _____
14. GCF of 8 and 12 = _____
15. GCF of 15 and 25 = _____
16. GCF of 18 and 24 = _____
17. LCM of 3 and 4 = _____
18. LCM of 4 and 6 = _____
19. LCM of 6 and 8 = _____
20. LCM of 5 and 10 = _____

TEST 7B • LEVEL 2: CORE FLUENCY Goal: 5 min

GCF, LCM & Order of Operations Without Parentheses

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. GCF of 24 and 36 = _____
2. GCF of 30 and 45 = _____
3. GCF of 32 and 48 = _____
4. GCF of 42 and 56 = _____
5. LCM of 8 and 12 = _____
6. LCM of 9 and 12 = _____
7. LCM of 12 and 15 = _____
8. LCM of 10 and 14 = _____
9. $10 + 5 \times 2 =$ _____
10. $25 - 5 \times 3 =$ _____
11. $4 + 6 \div 2 =$ _____
12. $12 - 2 \times 4 =$ _____
13. $15 - 3 \times 3 =$ _____
14. $30 - 4 \times 5 =$ _____
15. $2^3 \times 3^2 =$ _____
16. $2^2 \times 3 \times 5 =$ _____
17. $2^3 \times 5^2 =$ _____
18. $3^2 + 4^2 =$ _____
19. $5^2 - 3^2 =$ _____
20. $10^2 - 8^2 =$ _____

Section 7: Primes, GCF/LCM & Nested Expressions

Prime factorization, GCF, LCM, powers, and multi-branch expression trees. • Multi-Step Speed (Test 7C) & Advanced Challenge (Test 7D)

TEST 7C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Nested Two-Branch Expression Trees

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $(8 + 2) \times (9 - 4) =$ _____
2. $((14 - 6) \times 3) + (24 \div 4) =$ _____
3. $(30 - (4 \times 5)) \times (12 \div (2 + 4)) =$ _____
4. $((5 + 3) \times (8 - 2)) \div (6 - 2) =$ _____
5. $((18 \div 3) + 4) \times ((15 - 7) \div 2) =$ _____
6. $(10 + 5 \times 2) \div (12 - 2 \times 4) =$ _____
7. $(25 - 5 \times 3) \times (4 + 6 \div 2) =$ _____
8. $((7 + 3) \times 4 - 8) \div 4 =$ _____
9. $(48 \div (2 + 6)) \times (15 - 3 \times 3) =$ _____
10. $(16 - (4 + 2)) \times (18 \div (3 \times 2)) =$ _____
11. $(3^2 + 1) \times (4^2 - 11) =$ _____
12. $(5^2 - 5) \div (2^3 - 4) =$ _____
13. $((20 - 5) \div 3) \times (12 - 4) =$ _____
14. $(36 \div (3 \times 3)) + (5 \times 4) =$ _____
15. $(50 - 2 \times 15) \div (3 + 2) =$ _____
16. $((9 + 3) \times 5) \div (10 - 4) =$ _____
17. $(2^4 + 4) \times (3^2 - 6) =$ _____
18. $(100 - 6^2) \div (2^3) =$ _____
19. $(7^2 - 9) \div (2^2 + 4) =$ _____
20. $((12 + 8) \div 4)^3 =$ _____

TEST 7D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Number Theory: GCF × LCM, Divisor Counts & Deep Trees

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $\text{GCF}(12, 18) \times \text{LCM}(12, 18) =$ _____
2. $\text{GCF}(15, 20) \times \text{LCM}(15, 20) =$ _____
3. GCF of 72, 108, and 144 = _____
4. LCM of 6, 8, and 12 = _____
5. LCM of 4, 9, and 12 = _____
6. LCM of 5, 6, and 8 = _____
7. Number of positive factors of 12 = _____
8. Number of positive factors of 24 = _____
9. Number of positive factors of 36 = _____
10. Sum of prime factors of 30 (2+3+5) = _____
11. Sum of all positive factors of 12 = _____
12. $(6^2 - 4^2) \div (6 - 4) =$ _____
13. $(15^2 - 5^2) \div (15 + 5) =$ _____
14. $(25^2 - 15^2) \div (25 - 15) =$ _____
15. $((35 - 7) \times 4) \div (14 + 2) =$ _____
16. $(2^5 - 2^3) \div (3^2 - 3) =$ _____
17. $(3^4 - 1) \div (4^2 + 0) =$ _____
18. $(10^3 - 200) \div (4^2 \times 5) =$ _____
19. $((8^2 - 4) \div 15) \times (5^2 - 10) =$ _____
20. $(12^2 + 5^2) \div 13 =$ _____

Section 8: Algebra, Quadratics & Coordinate Geometry

Linear equations, slope, Pythagorean distance, and quadratic roots. • Study the Champion Way & complete W1–W6 before starting Tests 8A–8D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 8: ALGEBRA, QUADRATICS & COORDINATE GEOMETRY

RULE 1 **Rule 1 — Vieta's Root Formulas (Zero Factoring Required!): For $ax^2 + bx + c = 0$, the Sum of Roots is $-b/a$ and the Product of Roots is c/a. Never solve for individual roots first!**

RULE 2 Rule 2 — Horner's Nested Form for Polynomials: Evaluate $ax^2 + bx + c$ as $(ax + b)x + c$ so you only type x without squaring separately (or use [POLY] on SC-32 for instant roots).

RULE 3 Rule 3 — Pythagorean Triples & 3D Space Diagonals: Memorize (3,4,5), (5,12,13), (8,15,17), (7,24,25), (9,40,41) and 3D triples (3,4,12 → 13), (2,3,6 → 7), (1,4,8 → 9).

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — Vieta's Sum & Product of Roots: For $2x^2 - 10x + 7 = 0$, find Sum of Roots Stack Keys: `1` `0` `ENTER` `2` `÷`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Apply the quadratic formula $(-b \pm \sqrt{(b^2 - 4ac)}) / 2a$ and add the two radicals.

CHAMPION SPEED WAY (Use This!):

Champion Way: Sum of roots = -b / a = -(-10) / 2 = 5 in 1 second!

Archetype 2 — 2×2 Linear System by Elimination: Solve $x + y = 10$ and $x - y = 4$ Stack Keys: `1` `0` `ENTER` `4` `+` `2` `÷`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Substitute $y = 10 - x$ into $x - (10 - x) = 4$ and solve.

CHAMPION SPEED WAY (Use This!):

Champion Way: For $x + y = S$ and $x - y = D$, x is the average $(S + D) \div 2 = 14 \div 2 = 7$, and y is $(S - D) \div 2 = 3$!

Archetype 3 — 3D Box Diagonal Pipeline: Evaluate $\sqrt{(3^2 + 4^2 + 12^2)}$ Stack Keys: `3` `↶` `x²` `4` `↶` `x²` `+` `1` `2` `↶` `x²` `+` `√x`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Compute $9 + 16 + 144 = 169$ on paper and take $\sqrt{169}$.

CHAMPION SPEED WAY (Use This!):

Champion Way: Spot chained triples (3,4 → 5, then 5,12 → 13), or key [3][x²][4][x²][+][12][x²][+][√x]!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. **Sum of roots of $x^2 - 9x + 20 = 0$:** _____

Shortcut Cue: Vieta's rule: -b / a = 9 / 1

W2. **Product of roots of $x^2 + 11x + 24 = 0$:** _____

Shortcut Cue: Vieta's rule: c / a = 24 / 1

W3. **Vertex x-coord of $y = x^2 - 8x + 19$:** _____

Shortcut Cue: Axis of symmetry x = -b / (2a) = 8 / 2

W4. **Solve for x: $x + y = 12$, $x - y = 6$:** _____

Shortcut Cue: Average (12 + 6) ÷ 2

W5. **Hypotenuse for legs 7 and 24 =** _____

Shortcut Cue: Pythagorean triple (7, 24, 25)

W6. **3D diagonal $\sqrt{(2^2 + 3^2 + 6^2)}$ =** _____

Shortcut Cue: $\sqrt{(4 + 9 + 36)} = \sqrt{49}$

Warm-Up Self-Check Answers: **W1: 9** | **W2: 24** | **W3: 4** | **W4: 9** | **W5: 25** | **W6: 7**

Section 8: Algebra, Quadratics & Coordinate Geometry

Linear equations, slope, Pythagorean distance, and quadratic roots. • Foundation (Test 8A) & Core Fluency (Test 8B)

TEST 8A • LEVEL 1: FOUNDATION Goal: 4 min

Evaluating Expressions & Solving Linear Equations

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. **3x + 5 when x = 4 :** _____
2. **4x - 7 when x = 5 :** _____
3. **$2x^2$ when x = 3 :** _____
4. **$x^2 + 4x$ when x = 2 :** _____
5. **5a - 2b when a = 4, b = 3 :** _____
6. **Solve: x + 9 = 21** _____
7. **Solve: x - 8 = 15** _____
8. **Solve: 4x = 36** _____
9. **Solve: x ÷ 5 = 7** _____
10. **Solve: 2x + 3 = 15** _____
11. **Solve: 3x - 4 = 14** _____
12. **Solve: 5x + 10 = 45** _____
13. **Solve: 4x - 9 = 11** _____
14. **Solve: 6x + 2 = 32** _____
15. **Solve: (x ÷ 3) + 2 = 6** _____
16. **Evaluate -8 + 15 =** _____
17. **Evaluate 6 - 14 =** _____
18. **Evaluate (-4) × (-6) =** _____
19. **Evaluate (-36) ÷ 9 =** _____
20. **Evaluate $(-5)^2$ =** _____

TEST 8B • LEVEL 2: CORE FLUENCY Goal: 5 min

Slope, Midpoints & Pythagorean Right Triangles

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. **Slope from (0, 0) to (4, 12) =** _____
2. **Slope from (1, 2) to (5, 10) =** _____
3. **Slope from (2, 3) to (6, 15) =** _____
4. **Slope from (-1, -2) to (3, 10) =** _____
5. **Slope from (0, 10) to (5, 0) =** _____
6. **Slope from (2, 8) to (6, 6) =** _____
7. **Midpoint x-value of (2, 5) & (8, 9) =** _____
8. **Midpoint y-value of (2, 5) & (8, 9) =** _____
9. **Hypotenuse for legs 3 and 4 =** _____
10. **Hypotenuse for legs 6 and 8 =** _____
11. **Hypotenuse for legs 5 and 12 =** _____
12. **Hypotenuse for legs 8 and 15 =** _____
13. **Hypotenuse for legs 9 and 12 =** _____
14. **Hypotenuse for legs 12 and 16 =** _____
15. **Missing leg: hyp = 10, leg = 6 →** _____
16. **Missing leg: hyp = 13, leg = 5 →** _____
17. **Distance from (0, 0) to (3, 4) =** _____
18. **Distance from (1, 1) to (4, 5) =** _____
19. **Distance from (3, 4) to (9, 12) =** _____
20. **Distance from (-3, 5) to (5, -10) =** _____

Section 8: Algebra, Quadratics & Coordinate Geometry

Linear equations, slope, Pythagorean distance, and quadratic roots. • Multi-Step Speed (Test 8C) & Advanced Challenge (Test 8D)

TEST 8C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Polynomial Evaluation & Quadratic Factoring

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Evaluate $x^2 + 5x + 6$ at $x = 2$: _____
2. Evaluate $2x^2 + 4x - 5$ at $x = 3$: _____
3. Evaluate $x^2 - 6x + 9$ at $x = 5$: _____
4. Evaluate $3x^2 - 2x + 1$ at $x = -2$: _____
5. Greater root of $x^2 + 5x + 6 = 0$: _____
6. Lesser root of $x^2 + 5x + 6 = 0$: _____
7. Greater root of $x^2 + 7x + 12 = 0$: _____
8. Greater root of $x^2 + 8x + 15 = 0$: _____
9. Positive root of $x^2 - x - 6 = 0$: _____
10. Lesser root of $x^2 - 5x + 6 = 0$: _____
11. Greater root of $x^2 - 5x + 6 = 0$: _____
12. Positive root of $x^2 - 7x + 10 = 0$ (max) : _____
13. Positive root of $x^2 - 9 = 0$: _____
14. Positive root of $x^2 - 16 = 0$: _____
15. Positive root of $x^2 - 25 = 0$: _____
16. Repeated root of $x^2 + 10x + 25 = 0$: _____
17. Repeated root of $x^2 - 8x + 16 = 0$: _____
18. Greater root of $2x^2 + 5x + 2 = 0$: _____
19. Positive root of $x^2 + 2x - 15 = 0$: _____
20. Positive root of $x^2 - 3x - 10 = 0$: _____

TEST 8D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Vieta's Root Sums/Products, Linear Systems & 3D Distance

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Sum of roots of $x^2 - 9x + 20 = 0$: _____
2. Product of roots of $x^2 - 9x + 20 = 0$: _____
3. Sum of roots of $x^2 + 11x + 24 = 0$: _____
4. Product of roots of $x^2 + 11x + 24 = 0$: _____
5. Sum of roots of $2x^2 - 10x + 7 = 0$: _____
6. Product of roots of $2x^2 - 10x + 7 = 0$: _____
7. Discriminant $b^2 - 4ac$ of $x^2 + 6x + 5$: _____
8. Discriminant $b^2 - 4ac$ of $x^2 - 4x + 4$: _____
9. Vertex x-coord of $y = x^2 - 8x + 19$: _____
10. Minimum value of $y = x^2 - 8x + 19$: _____
11. Solve for x: $x + y = 10$, $x - y = 4$: _____
12. Solve for y: $x + y = 10$, $x - y = 4$: _____
13. Solve for x: $2x + y = 13$, $x + y = 8$: _____
14. Solve for x: $3x - 4 = x + 10$: _____
15. Solve for x: $5x + 2 = 2x + 17$: _____
16. Hypotenuse for legs 7 and 24 = _____
17. Hypotenuse for legs 9 and 40 = _____
18. 3D diagonal $\sqrt{(3^2 + 4^2 + 12^2)}$ = _____
19. 3D diagonal $\sqrt{(2^2 + 3^2 + 6^2)}$ = _____
20. 3D diagonal $\sqrt{(1^2 + 4^2 + 8^2)}$ = _____

Cumulative Review C2: Intermediate Math (Sections 1–8)

Comprehensive review across fractions, expressions, number theory, and algebra. • Study the Champion Way & complete W1–W6 before starting Tests C2A–C2D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — CUMULATIVE REVIEW C2: INTERMEDIATE MATH (SECTIONS 1–8)

RULE 1	**Rule 1 — Match the Fastest Tool to the Problem: Use mental identities for (a^2 - b^2), Vieta's formulas, and GCF×LCM, and use inside-out RPN chains for multi-branch arithmetic.**
RULE 2	Rule 2 — Clear Fractions Before Solving Equations: Multiply or distribute the LCD first to turn fraction expressions into 1-step integer subtraction.
RULE 3	Rule 3 — Never Re-Type Intermediate Results: Keep intermediate values on Line Y (Waiting Shelf) while computing the second branch on Line X.

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — Euclidean Distance Between Points: Find distance from (-3, 5) to (5, -10) Stack Keys: [8] [ENTER] [1] [5] [↶] [▸θ,r]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Write d = $\sqrt{((5 - (-3))^2 + (-10 - 5)^2)}$ on paper and compute step-by-step.

CHAMPION SPEED WAY (Use This!):

Champion Way: Find dx = 8 and dy = 15 mentally, then spot the (8, 15, 17) triple (or press [8] [ENTER] [15] [→ϑ,r] → 17)!

Archetype 2 — Parallel Fraction Reciprocal: Evaluate 1 ÷ (1/4 + 1/6) Stack Keys: [4] [x^{-1}] [6] [x^{-1}] [+] [x^{-1}]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Common denominator 12 → 5/12 → 12/5 = 2.4.

CHAMPION SPEED WAY (Use This!):

Champion Way: Key [4] [1/x] [6] [1/x] [+] [1/x] → 2.4 in 6 keys!

Archetype 3 — Continued Fraction Evaluation: Evaluate 1 + 1/(1 + 1/2) Stack Keys: [1] [ENTER] [•] [5] [+] [x^{-1}] [1] [+]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Convert layers on paper.

CHAMPION SPEED WAY (Use This!):

Champion Way: Start at bottom 1 + 1/2 = 3/2, invert to 2/3, add 1 → 5/3!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. **3/4 + (1/2 × 1/4) (decimal) =** _____

Shortcut Cue: 0.75 + 0.125

W2. **(2/3 + 3/4) × 12 =** _____

Shortcut Cue: Distribute 12: 8 + 9

W3. **Distance from (3, 4) to (9, 12) =** _____

Shortcut Cue: dx = 6, dy = 8 → (6, 8, 10) triple

W4. **Sum of roots of $2x^2$ - 14x + 9 = 0 :** _____

Shortcut Cue: Vieta's -b / a = 14 / 2

W5. **Minimum value of $y = x^2$ - 6x + 14 :** _____

Shortcut Cue: $(x - 3)^2$ + 5 → min is 5

W6. **2/3 × 3/4 × 4/5 × 5/6 =** _____

Shortcut Cue: Telescoping cancel → 2/6 = 1/3

Warm-Up Self-Check Answers: **W1: 0.875** | **W2: 17** | **W3: 10** | **W4: 7** | **W5: 5** | **W6: 1/3**

Cumulative Review C2: Intermediate Math (Sections 1–8)

Comprehensive review across fractions, expressions, number theory, and algebra. • Foundation (Test C2A) & Core Fluency (Test C2B)

TEST C2A • LEVEL 1: FOUNDATION Goal: 4 min

Multi-Step Arithmetic, Fractions & Basic Algebra

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	18 + 18 - 9 =	_____
2.	100 - 36 =	_____
3.	(18 - 6) × 3 =	_____
4.	(24 + 16) ÷ 8 =	_____
5.	1/2 + 1/4 (decimal) =	_____
6.	3/4 - 1/2 (decimal) =	_____
7.	2/3 of 18 =	_____
8.	3/4 of 24 =	_____
9.	$2^3 \times 3$ =	_____
10.	GCF of 12 and 18 =	_____
11.	LCM of 4 and 6 =	_____
12.	10 + 5 × 2 =	_____
13.	30 - 4 × 5 =	_____
14.	Solve: x + 12 = 30	_____
15.	Solve: 3x = 42	_____
16.	Solve: 2x + 5 = 19	_____
17.	Evaluate 3x - 4 at x = 6 :	_____
18.	Hypotenuse for legs 3 and 4 =	_____
19.	Hypotenuse for legs 6 and 8 =	_____
20.	Slope from (0, 0) to (3, 12) =	_____

TEST C2B • LEVEL 2: CORE FLUENCY Goal: 5 min

Rational Operations, Expression Trees & Coordinate Slope

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	3 1/2 - 1 3/4 (decimal) =	_____
2.	2 1/2 + 1 1/4 (decimal) =	_____
3.	Reciprocal of 4/5 (decimal) =	_____
4.	3/4 ÷ 1/2 (decimal) =	_____
5.	((32 - 14) × 5) ÷ 6 =	_____
6.	((15 + 5) × (8 - 3)) ÷ 10 =	_____
7.	80 - (15 × 4) + 12 =	_____
8.	(8 + 2) × (9 - 4) =	_____
9.	GCF of 24 and 36 =	_____
10.	LCM of 8 and 12 =	_____
11.	Slope from (-1, -2) to (3, 10) =	_____
12.	Distance from (0, 0) to (12, 16) =	_____
13.	Distance from (3, 4) to (9, 12) =	_____
14.	Positive root of $x^2 - 9 = 0$:	_____
15.	Positive root of $x^2 - 16 = 0$:	_____
16.	Greater root of $x^2 + 5x + 6 = 0$:	_____
17.	Lesser root of $x^2 - 5x + 6 = 0$:	_____
18.	Evaluate $x^2 + 3x$ at x = 4 :	_____
19.	15^2 =	_____
20.	25^2 =	_____

Cumulative Review C2: Intermediate Math (Sections 1–8)

Comprehensive review across fractions, expressions, number theory, and algebra. • Multi-Step Speed (Test C2C) & Advanced Challenge (Test C2D)

TEST C2C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Complex Fractions, Quadratics & Multi-Branch Evaluation

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $3/4 + (1/2 \times 1/4)$ (decimal) = _____
2. $1 \div (1/4 + 1/6)$ (decimal) = _____
3. $(1/3 + 1/4) \times 24$ = _____
4. $(2/3 + 3/4) \times 12$ = _____
5. $((5 + 3) \times (8 - 2)) \div (6 - 2)$ = _____
6. $((18 \div 3) + 4) \times ((15 - 7) \div 2)$ = _____
7. $(25 - 5 \times 3) \times (4 + 6 \div 2)$ = _____
8. Evaluate $2x^2 + 4x - 5$ at $x = 3$: _____
9. Positive root of $x^2 - x - 6 = 0$: _____
10. Greater root of $2x^2 + 5x + 2 = 0$: _____
11. Repeated root of $x^2 + 10x + 25 = 0$: _____
12. Sum of roots of $x^2 - 7x + 12 = 0$: _____
13. Product of roots of $x^2 - 7x + 12 = 0$: _____
14. Solve: $4x - 5 = 2x + 9$ _____
15. Missing leg: hyp = 13, leg = 5 → _____
16. Missing leg: hyp = 17, leg = 8 → _____
17. Number of positive factors of 24 = _____
18. LCM of 6, 8, and 12 = _____
19. 19×21 = _____
20. 24×26 = _____

TEST C2D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Competition Number Sense, Algebra & Geometry

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $1 + 2 + 3 + ... + 10$ = _____
2. $1 + 3 + 5 + 7 + 9 + 11$ = _____
3. 35^2 = _____
4. $25^2 - 15^2$ = _____
5. $24 \times (5/6 - 3/8)$ = _____
6. $1 + 1/(1 + 1/2)$ = _____
7. GCF(15, 20) × LCM(15, 20) = _____
8. Sum of positive factors of 12 = _____
9. $((35 - 7) \times 4) \div (14 + 2)$ = _____
10. $(12^2 + 5^2) \div 13$ = _____
11. Distance from (-3, 5) to (5, -10) = _____
12. 3D diagonal $\sqrt{(3^2 + 4^2 + 12^2)}$ = _____
13. Solve for x: $x + y = 12$, $x - y = 6$: _____
14. Minimum value of $y = x^2 - 6x + 14$: _____
15. Sum of roots of $2x^2 - 14x + 9 = 0$: _____
16. Hypotenuse for legs 7 and 24 = _____
17. $(45 \times 12) \div 18$ = _____
18. $1000 - 384$ = _____
19. $99 + 98 + 97$ = _____
20. $2/3 \times 3/4 \times 4/5 \times 5/6$ = _____

Section 9: Trigonometry, Vectors & Polar Coordinates

Special angles, right-triangle projections, inverse trig, and vector magnitudes. • Study the Champion Way & complete W1–W6 before starting Tests 9A–9D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 9: TRIGONOMETRY, VECTORS & POLAR COORDINATES

RULE 1	**Rule 1 — The 3-Key Vector & Hypotenuse One-Liner ([→ϑ,r]): Never type √(x² + y²) or arctan(y/x) separately! Enter x [ENTER] y [yellow] [→ϑ,r] to get magnitude r in X and angle ϑ in Y!**
RULE 2	Rule 2 — Maximum of a·sin(x) + b·cos(x) Is √(a² + b²): The amplitude (maximum value) of a·sin(x) + b·cos(x) is simply the hypotenuse √(a² + b²) (e.g., max of 5 sin x + 12 cos x is 13!).
RULE 3	Rule 3 — Pythagorean & Double-Angle Trig Identities: Spot $\sin^2\vartheta + \cos^2\vartheta = 1$, $\sec^2\vartheta - \tan^2\vartheta = 1$, and $2\sin(\vartheta)\cos(\vartheta) = \sin(2\vartheta)$ before touching the calculator!

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — Polar Magnitude & Angle One-Liner: Find magnitude and angle of <3, 4> Stack Keys: 3 ENTER 4 ↶ ▸θ,r

SLOW / AMATEUR WAY (Avoid):

Slow Way: Compute √(3² + 4²) = 5, then clear and compute atan(4 ÷ 3) = 53.13°.

CHAMPION SPEED WAY (Use This!):

Champion Way: Key [3] [ENTER] [4] [yellow] [→ϑ,r] → Line X shows r = 5, Line Y shows ϑ = 53.13° simultaneously!

Archetype 2 — Linear Trig Combination: Evaluate 12 cos(45°) + 5 sin(45°) Stack Keys: 1 7 ENTER 4 5 SIN ×

SLOW / AMATEUR WAY (Avoid):

Slow Way: Compute 12 × cos(45°), write it down, compute 5 × sin(45°), add.

CHAMPION SPEED WAY (Use This!):

Champion Way: Since cos(45°) = sin(45°), factor it out: (12 + 5) × sin(45°) = 17 × sin(45°) = 12.0208 in 5 keys!

Archetype 3 — Wave Amplitude Shortcut: Find max value of 8 sin(x) - 15 cos(x) Stack Keys: 8 ENTER 1 5 ↶ ▸θ,r

SLOW / AMATEUR WAY (Avoid):

Slow Way: Take derivatives or test angles.

CHAMPION SPEED WAY (Use This!):

Champion Way: Max of a sin(x) ± b cos(x) is √(a² + b²) = √(8² + 15²) = 17!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. Vector <5, 12> → Magnitude r = _____

Shortcut Cue: Pythagorean triple or [5][ENTER][12][→ϑ,r]

W2. $5\sin^2(20°) + 5\cos^2(20°)$ = _____

Shortcut Cue: Factor $5(\sin^2 + \cos^2) = 5(1)$

W3. Area of Δ: a = 8, b = 10, C = 30° → _____

Shortcut Cue: 1/2 × a × b × sin(30°) = 0.5 × 80 × 0.5

W4. 2 sin(15°) cos(15°) = _____

Shortcut Cue: Double-angle identity: sin(30°)

W5. max value of 3 sin(x) + 4 cos(x) = _____

Shortcut Cue: Hypotenuse √(3² + 4²)

W6. Law of Cosines: a=8, b=5, C=60° → c = _____

Shortcut Cue: $c^2 = 64 + 25 - 2(8)(5)(0.5) = 49 \rightarrow c = 7$

Warm-Up Self-Check Answers: W1: 13 | W2: 5 | W3: 20 | W4: 0.5 | W5: 5 | W6: 7

Section 9: Trigonometry, Vectors & Polar Coordinates

Special angles, right-triangle projections, inverse trig, and vector magnitudes. • Foundation (Test 9A) & Core Fluency (Test 9B)

TEST 9A • LEVEL 1: FOUNDATION Goal: 4 min

Exact Special-Angle Trig Ratios & Degree/Radian Conversions

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	$\sin(0°) =$	_____
2.	$\cos(0°) =$	_____
3.	$\sin(30°) =$	_____
4.	$\cos(60°) =$	_____
5.	$\sin(90°) =$	_____
6.	$\cos(90°) =$	_____
7.	$\tan(0°) =$	_____
8.	$\tan(45°) =$	_____
9.	$2 \times \sin(30°) =$	_____
10.	$4 \times \cos(60°) =$	_____
11.	$\sin(30°) + \cos(60°) =$	_____
12.	$5 \times \tan(45°) =$	_____
13.	Acute angle ϑ (deg): $\sin(\vartheta) = 0.5 \rightarrow$	_____
14.	Acute angle ϑ (deg): $\cos(\vartheta) = 0.5 \rightarrow$	_____
15.	Acute angle ϑ (deg): $\tan(\vartheta) = 1 \rightarrow$	_____
16.	Right triangle: $90° - 35° =$	_____
17.	Right triangle: $90° - 62° =$	_____
18.	$180°$ in radians (multiple of π) =	_____
19.	$90°$ in radians =	_____
20.	$60°$ in radians =	_____

TEST 9B • LEVEL 2: CORE FLUENCY Goal: 5 min

Right-Triangle Projections & Vector Magnitudes

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	Hyp = 10, $\vartheta = 30° \rightarrow$ Opp leg y =	_____
2.	Hyp = 12, $\vartheta = 60° \rightarrow$ Adj leg x =	_____
3.	Hyp = 20, $\vartheta = 30° \rightarrow$ Opp leg y =	_____
4.	Hyp = 16, $\vartheta = 60° \rightarrow$ Adj leg x =	_____
5.	Adj = 9, $\vartheta = 45° \rightarrow$ Opp leg y =	_____
6.	Opp = 7, $\vartheta = 30° \rightarrow$ Hypotenuse r =	_____
7.	Adj = 5, $\vartheta = 60° \rightarrow$ Hypotenuse r =	_____
8.	Polar ($r = 10$, $\vartheta = 0°$) $\rightarrow$ x =	_____
9.	Polar ($r = 10$, $\vartheta = 0°$) $\rightarrow$ y =	_____
10.	Polar ($r = 8$, $\vartheta = 90°$) $\rightarrow$ y =	_____
11.	Polar ($r = 14$, $\vartheta = 60°$) $\rightarrow$ x =	_____
12.	Polar ($r = 18$, $\vartheta = 30°$) $\rightarrow$ y =	_____
13.	Vector $\langle 3, 4 \rangle \rightarrow$ Magnitude r =	_____
14.	Vector $\langle 6, 8 \rangle \rightarrow$ Magnitude r =	_____
15.	Vector $\langle 5, 12 \rangle \rightarrow$ Magnitude r =	_____
16.	Vector $\langle 8, 6 \rangle \rightarrow$ Magnitude r =	_____
17.	Vector $\langle 9, 12 \rangle \rightarrow$ Magnitude r =	_____
18.	Vector $\langle 8, 15 \rangle \rightarrow$ Magnitude r =	_____
19.	Vector $\langle 12, 16 \rangle \rightarrow$ Magnitude r =	_____
20.	Vector $\langle 7, 24 \rangle \rightarrow$ Magnitude r =	_____

Section 9: Trigonometry, Vectors & Polar Coordinates

Special angles, right-triangle projections, inverse trig, and vector magnitudes. • Multi-Step Speed (Test 9C) & Advanced Challenge (Test 9D)

TEST 9C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Trigonometric Identities, Squares & Vector Operations

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	$\sin^2(30°) =$	_____
2.	$\cos^2(30°) =$	_____
3.	$\sin^2(45°) =$	_____
4.	$\cos^2(45°) =$	_____
5.	$\tan^2(60°) =$	_____
6.	$\sin^2(37°) + \cos^2(37°) =$	_____
7.	$5 \sin^2(20°) + 5 \cos^2(20°) =$	_____
8.	$4 \cos^2(30°) =$	_____
9.	$8 \sin^2(45°) =$	_____
10.	$2 \sin(30°) \cos(60°) =$	_____
11.	$\langle 2, 3\rangle + \langle 1, 1\rangle \rightarrow$ Magnitude r =	_____
12.	$\langle 4, 1\rangle + \langle 2, 7\rangle \rightarrow$ Magnitude r =	_____
13.	$\langle 8, 5\rangle - \langle 3, -7\rangle \rightarrow$ Magnitude r =	_____
14.	Dot product $\langle 2, 3\rangle \cdot \langle 4, 5\rangle =$	_____
15.	Dot product $\langle 3, -2\rangle \cdot \langle 4, 6\rangle =$	_____
16.	Area of Δ: a = 8, b = 10, C = 30° →	_____
17.	Area of Δ: a = 6, b = 12, C = 30° →	_____
18.	$12 \cos(60°) + 8 \sin(30°) =$	_____
19.	$20 \sin(30°) - 6 \tan(45°) =$	_____
20.	$\sec(60°) = 1 / \cos(60°) =$	_____

TEST 9D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

4-Sig-Fig Calculator Projections, Polar Angles & Law of Cosines

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	$12 \cos(45°) + 5 \sin(45°)$ (4 dec) =	_____
2.	$10 \cos(30°)$ (4 dec) =	_____
3.	$10 \sin(45°)$ (4 dec) =	_____
4.	$10 \tan(30°)$ (4 dec) =	_____
5.	Vector $\langle 3, 4\rangle$ angle ϑ (deg, 2 dec) =	_____
6.	Vector $\langle 4, 3\rangle$ angle ϑ (deg, 2 dec) =	_____
7.	Vector $\langle 5, 5\rangle$ angle ϑ (deg) =	_____
8.	Vector $\langle 0, -10\rangle$ angle ϑ (deg) =	_____
9.	Law of Cosines: a=3, b=5, C=60° → c^2 =	_____
10.	Law of Cosines: a=8, b=5, C=60° → c =	_____
11.	Law of Sines: a=10, A=30°, B=90° → b =	_____
12.	$2 \sin(15°) \cos(15°) = \sin(30°) =$	_____
13.	$\cos^2(15°) - \sin^2(15°) = \cos(30°)$ (4 dec) =	_____
14.	$1 + \tan^2(45°) = \sec^2(45°) =$	_____
15.	$\sec^2(30°) - \tan^2(30°) =$	_____
16.	max value of $3 \sin(x) + 4 \cos(x) =$	_____
17.	max value of $5 \sin(x) + 12 \cos(x) =$	_____
18.	max value of $8 \sin(x) - 15 \cos(x) =$	_____
19.	Period of sin(2x) in degrees =	_____
20.	Period of cos(4x) in degrees =	_____

Section 10: Exponents, Logarithms & Growth Models

Powers, fractional exponents, base-10 and natural logarithms, and exponential growth. • Study the Champion Way & complete W1–W6 before starting Tests 10A–10D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 10: EXPONENTS, LOGARITHMS & GROWTH MODELS

RULE 1	**Rule 1 — Inside-Out Growth & Compounding Pipeline: For $P(1 + r/n)^{nt}$ or $P \cdot e^{rt}$, ALWAYS compute the inner rate or exponent first, apply [y^x] or [e^x], and multiply by P last!**
RULE 2	Rule 2 — Change-of-Base Logarithm Shortcut: To evaluate $\log_b(a)$, recognize exact powers $b^k = a$ (so $\log_4(64) = 3$), or key [a] [LN] [b] [LN] [÷] on the SC-32.
RULE 3	Rule 3 — Fractional Exponents Root-First Rule: To evaluate $a^{m/n}$ (like $27^{2/3}$ or $16^{3/4}$), always take the nth root FIRST to keep numbers small, then raise to power m ($27^{1/3} = 3 \rightarrow 3^2 = 9$)!

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — Continuous Growth Pipeline: Evaluate $500 \times e^{0.04 \times 5}$ — Stack Keys: [•] [0] [4] [ENTER] [5] [×] [e^x] [5] [0] [0] [×]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Type 500 first, then struggle to compute $e^{0.04 \times 5}$ without losing 500.

CHAMPION SPEED WAY (Use This!):

Champion Way: Start at the exponent: [.04] [ENTER] [5] [×] [e^x] [500] [×] → 610.7014 in 7 keys!

Archetype 2 — Monthly Compound Interest: Evaluate $250 \times (1 + 0.06 \div 12)^{24}$ — Stack Keys: [•] [0] [6] [ENTER] [1] [2] [÷] [1] [+] [2] [4] [y^x] [2] [5] [0] [×]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Compute 0.06 ÷ 12 on paper, add 1, raise to 24, multiply by 250.

CHAMPION SPEED WAY (Use This!):

Champion Way: Inside-out chain: [.06] [ENTER] [12] [÷] [1] [+] [24] [y^x] [250] [×] → 281.79!

Archetype 3 — Fractional Exponent Root-First: Evaluate $64^{2/3}$ — Stack Keys: [6] [4] [ENTER] [3] [x^{-1}] [2] [×] [y^x]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Square $64^2 = 4096$ first, then try to find the cube root of 4096.

CHAMPION SPEED WAY (Use This!):

Champion Way: Take the cube root of 64 first ($\sqrt[3]{64} = 4$), then square: $4^2 = 16$!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. $\log_{10}(4) + \log_{10}(25) =$ _____

Shortcut Cue: $\log_{10}(4 \times 25) = \log_{10}(100)$

W2. Half-life (3 periods): $100 \times (0.5)^3 =$ _____

Shortcut Cue: $100 \div 8 = 12.5$

W3. $27^{2/3} =$ _____

Shortcut Cue: Cube root first (3), then square (3^2)

W4. $16^{3/4} =$ _____

Shortcut Cue: 4th root first (2), then cube (2^3)

W5. Change of base: $\ln(125) \div \ln(5) =$ _____

Shortcut Cue: $\log_5(125) = 3$ since $5^3 = 125$

W6. Solve: $4^x = 32$ (decimal) = _____

Shortcut Cue: Base 2: $2^{2x} = 2^5 \rightarrow 2x = 5$

Warm-Up Self-Check Answers: W1: 2 | W2: 12.5 | W3: 9 | W4: 8 | W5: 3 | W6: 2.5

Section 10: Exponents, Logarithms & Growth Models

Powers, fractional exponents, base-10 and natural logarithms, and exponential growth. • Foundation (Test 10A) & Core Fluency (Test 10B)

TEST 10A • LEVEL 1: FOUNDATION Goal: 4 min

Integer Powers, Zero/Negative Exponents & Roots

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $2^0 =$ _____
2. $e^0 =$ _____
3. $2^4 =$ _____
4. $2^5 =$ _____
5. $2^6 =$ _____
6. $2^8 =$ _____
7. $2^{10} =$ _____
8. $3^3 =$ _____
9. $3^4 =$ _____
10. $5^3 =$ _____
11. $5^4 =$ _____
12. $10^3 =$ _____
13. $10^4 =$ _____
14. 2^{-1} (decimal) $=$ _____
15. 2^{-2} (decimal) $=$ _____
16. 5^{-1} (decimal) $=$ _____
17. 10^{-2} (decimal) $=$ _____
18. $16^{1/2} =$ _____
19. $27^{1/3} =$ _____
20. $8^{2/3} =$ _____

TEST 10B • LEVEL 2: CORE FLUENCY Goal: 5 min

Base-10 Logs, Binary Logs & Natural Logarithm Identities

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $\log_{10}(1) =$ _____
2. $\log_{10}(10) =$ _____
3. $\log_{10}(100) =$ _____
4. $\log_{10}(1000) =$ _____
5. $\log_{10}(10000) =$ _____
6. $\log_{10}(0.1) =$ _____
7. $\log_{10}(0.01) =$ _____
8. $\ln(1) =$ _____
9. $\ln(e^1) =$ _____
10. $\ln(e^3) =$ _____
11. $\ln(e^{-2}) =$ _____
12. $e^{\ln 5} =$ _____
13. $e^{2 \ln 3} =$ _____
14. $10^{\log 7} =$ _____
15. $\log_2(8) =$ _____
16. $\log_2(32) =$ _____
17. $\log_2(64) =$ _____
18. $\log_3(27) =$ _____
19. $\log_5(125) =$ _____
20. $\log_{10}(2) + \log_{10}(5) =$ _____

Section 10: Exponents, Logarithms & Growth Models

Powers, fractional exponents, base-10 and natural logarithms, and exponential growth. • Multi-Step Speed (Test 10C) & Advanced Challenge (Test 10D)

TEST 10C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Logarithm Laws, Half-Life Decay & Compound Growth

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $\log_{10}(4) + \log_{10}(25) =$ _____
2. $\log_{10}(20) + \log_{10}(50) =$ _____
3. $\log_{10}(200) - \log_{10}(2) =$ _____
4. $\log_{10}(5000) - \log_{10}(5) =$ _____
5. $\log_{2}(24) - \log_{2}(3) =$ _____
6. $\log_{3}(18) - \log_{3}(2) =$ _____
7. Solve: $2^{x} = 64$ _____
8. Solve: $3^{x} = 81$ _____
9. Solve: $5^{x-1} = 25$ _____
10. Solve: $\log_{10}(x) = 3$ _____
11. Solve: $\log_{2}(x) = 5$ _____
12. Half-life (1 period): $100 \times (0.5)^{1} =$ _____
13. Half-life (2 periods): $100 \times (0.5)^{2} =$ _____
14. Half-life (3 periods): $100 \times (0.5)^{3} =$ _____
15. Half-life (4 periods): $160 \times (0.5)^{4} =$ _____
16. Doubling (3 periods): $50 \times 2^{3} =$ _____
17. Tripling (2 periods): $20 \times 3^{2} =$ _____
18. 1 year at 10%: $1000 \times 1.10 =$ _____
19. 2 years at 10%: $1000 \times (1.10)^{2} =$ _____
20. 2 years at 5%: $1000 \times (1.05)^{2} =$ _____

TEST 10D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Continuous Compounding, Doubling Time & Change of Base

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Doubling time: $\ln(2) \div 0.07$ (4 dec) = _____
2. Doubling time: $\ln(2) \div 0.05$ (4 dec) = _____
3. Continuous: $500 \times e^{0.04 \times 5}$ (4 dec) = _____
4. Continuous: $1500 \times e^{0.05 \times 10}$ (2 dec) = _____
5. Monthly: $250 \times (1 + 0.06 \div 12)^{24}$ (2 dec) = _____
6. 3 years at 10%: $1000 \times (1.10)^{3} =$ _____
7. Change of base: $\ln(8) \div \ln(2) =$ _____
8. Change of base: $\ln(81) \div \ln(3) =$ _____
9. Change of base: $\ln(125) \div \ln(5) =$ _____
10. Change of base: $\log_{10}(32) \div \log_{10}(2) =$ _____
11. $\log_{4}(64) =$ _____
12. $\log_{8}(64) =$ _____
13. $\log_{4}(8)$ (decimal) = _____
14. $\log_{9}(27)$ (decimal) = _____
15. $27^{2/3} =$ _____
16. $16^{3/4} =$ _____
17. $32^{3/5} =$ _____
18. $64^{2/3} =$ _____
19. Solve: $4^{x} = 32$ (decimal) = _____
20. Solve: $9^{x} = 27$ (decimal) = _____

Section 11: Combinatorics, Probability & Statistics

Factorials, permutations, combinations, probability ratios, and dataset statistics. • Study the Champion Way & complete W1–W6 before starting Tests 11A–11D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 11: COMBINATORICS, PROBABILITY & STATISTICS

RULE 1	**Rule 1 — Cancel Factorials Before Multiplying: Never compute 8! and 5! separately for 8C3! Cancel 5! immediately: 8C3 = (8 × 7 × 6) ÷ (3 × 2 × 1) = 8 × 7 = 56 (since 3×2 cancels 6)!**
RULE 2	Rule 2 — Pascal's Row Sum Identity: The sum of all combinations in Row n (nC0 + nC1 + ... + nCn) is always 2^n (so Row 5 sum = 2^5 = 32; Row 6 sum = 2^6 = 64).
RULE 3	Rule 3 — Symmetric Dataset SD & Mean Shortcuts: For three evenly spaced numbers {a - d, a, a + d}, the Mean is a and the Sample Standard Deviation is simply d!

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — Instant Combination Cancellation: Evaluate 8C3

Stack Keys: `8` `ENTER` `7` `×`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Compute 8! = 40,320, 3! = 6, 5! = 120, multiply 6 × 120 = 720, divide.

CHAMPION SPEED WAY (Use This!):

Champion Way: Write 3 descending factors (8 × 7 × 6) ÷ 6 → cancel 6 to get 8 × 7 = 56 in 1 second!

Archetype 2 — Word Arrangements with Repeated Letters: Arrangements of LEVEL

Stack Keys: `5` `↶` `x!` `4` `÷`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Write out permutations by hand.

CHAMPION SPEED WAY (Use This!):

Champion Way: 5 letters with two Ls (2!) and two Es (2!): 5! ÷ (2! × 2!) = 120 ÷ 4 = 30!

Archetype 3 — Evenly Spaced Sample Standard Deviation: Sample SD of {10, 14, 18}

Stack Keys: `1` `0` `Σ+` `1` `4` `Σ+` `1` `8` `Σ+`

SLOW / AMATEUR WAY (Avoid):

Slow Way: Compute deviations $(-4)^2$, 0^2, 4^2, sum to 32, divide by n-1 = 2 to get 16, take √16.

CHAMPION SPEED WAY (Use This!):

Champion Way: For any 3 evenly spaced numbers {a - d, a, a + d}, Sample SD = step d = 4 immediately!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. **7C2 =** _____

Shortcut Cue: (7 × 6) ÷ 2 = 7 × 3

W2. **8C3 =** _____

Shortcut Cue: (8 × 7 × 6) ÷ 6 = 8 × 7

W3. **Arrangements of BOOK (4! ÷ 2!) =** _____

Shortcut Cue: 24 ÷ 2

W4. **Row 5 sum: 5C0 + ... + 5C5 =** _____

Shortcut Cue: Pascal's Row Sum = 2^5

W5. **5 coins: P(exactly 2 Heads) =** _____

Shortcut Cue: 5C2 ÷ 2^5 = 10 ÷ 32

W6. **Sample SD of {2, 5, 8} =** _____

Shortcut Cue: 3 evenly spaced numbers → step d = 3

Warm-Up Self-Check Answers: **W1: 21** | **W2: 56** | **W3: 12** | **W4: 32** | **W5: 0.3125** | **W6: 3**

Section 11: Combinatorics, Probability & Statistics

Factorials, permutations, combinations, probability ratios, and dataset statistics. • Foundation (Test 11A) & Core Fluency (Test 11B)

TEST 11A • LEVEL 1: FOUNDATION Goal: 4 min

Factorials, Sample Means, Medians & Ranges

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	$0! =$	
2.	$1! =$	
3.	$2! =$	
4.	$3! =$	
5.	$4! =$	
6.	$5! =$	
7.	$6! =$	
8.	Sum: $1 + 2 + 3 + 4 + 5 =$	
9.	Mean of $\{2, 4, 6, 8, 10\} =$	
10.	Mean of $\{10, 20, 30\} =$	
11.	Mean of $\{4, 7, 10, 11\} =$	
12.	Median of $\{3, 5, 9, 12, 15\} =$	
13.	Median of $\{2, 6, 10, 14\} =$	
14.	Range of $\{4, 9, 15, 22\} =$	
15.	Mode of $\{2, 5, 5, 8, 9\} =$	
16.	Fair coin P(Heads) (decimal) $=$	
17.	Fair 6-sided die P(roll $= 6$) $=$	
18.	Fair 6-sided die P(even) (decimal) $=$	
19.	Fair 6-sided die P(roll > 4) $=$	
20.	3×4 Choices (Fundamental Rule) $=$	

TEST 11B • LEVEL 2: CORE FLUENCY Goal: 5 min

Permutations nPr, Combinations nCr & Factorial Quotients

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1.	$5! \div 4! =$	
2.	$6! \div 5! =$	
3.	$5P2 = 5! \div 3! =$	
4.	$6P2 = 6! \div 4! =$	
5.	$6P3 = 6! \div 3! =$	
6.	$7P2 = 7! \div 5! =$	
7.	$4C2 = 4! \div (2! \times 2!) =$	
8.	$5C2 = 5! \div (2! \times 3!) =$	
9.	$5C3 = 5! \div (3! \times 2!) =$	
10.	$6C2 = 6! \div (2! \times 4!) =$	
11.	$6C3 = 6! \div (3! \times 3!) =$	
12.	$7C2 = 7! \div (2! \times 5!) =$	
13.	$7C3 = 7! \div (3! \times 4!) =$	
14.	$8C2 = 8! \div (2! \times 6!) =$	
15.	$8C3 = 8! \div (3! \times 5!) =$	
16.	$9C2 = 9! \div (2! \times 7!) =$	
17.	$10C2 = 10! \div (2! \times 8!) =$	
18.	$10C3 = 10! \div (3! \times 7!) =$	
19.	$12C2 = 12! \div (2! \times 10!) =$	
20.	$100C1 =$	

Section 11: Combinatorics, Probability & Statistics

Factorials, permutations, combinations, probability ratios, and dataset statistics. • Multi-Step Speed (Test 11C) & Advanced Challenge (Test 11D)

TEST 11C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Independent Events, Arrangements & Weighted Averages

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. **2 coins: P(both Heads) (decimal) =** _____
2. **3 coins: P(all 3 Heads) (decimal) =** _____
3. **3 coins: P(at least 1 Tail) (decimal) =** _____
4. **2 dice: Total outcomes (6 × 6) =** _____
5. **2 dice: Ways to roll sum = 7 =** _____
6. **2 dice: P(sum = 7) =** _____
7. **2 dice: P(sum = 2) =** _____
8. **2 dice: P(sum = 12) =** _____
9. **Arrangements of letters in CAT (3!) =** _____
10. **Arrangements of MATH (4!) =** _____
11. **Arrangements of STACK (5!) =** _____
12. **Arrangements of MOM (3! ÷ 2!) =** _____
13. **Arrangements of BOOK (4! ÷ 2!) =** _____
14. **Arrangements of LEVEL (5! ÷ (2!2!)) =** _____
15. **Handshakes among 6 people (6C2) =** _____
16. **Handshakes among 8 people (8C2) =** _____
17. **Diagonals in a hexagon (6C2 - 6) =** _____
18. **Mean of 5 numbers is 12 → Sum =** _____
19. **Mean of {4, 8, 12, x} is 10 → x =** _____
20. **1 ÷ 6C2 (4 dec) =** _____

TEST 11D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Binomial Probability, Pascal's Triangle & Variance

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. **Row 4 sum: 4C0+4C1+4C2+4C3+4C4 =** _____
2. **Row 5 sum: 5C0 + ... + 5C5 = 2^5 =** _____
3. **Row 6 sum: 6C0 + ... + 6C6 = 2^6 =** _____
4. **4 coins: P(exactly 2 Heads) = 6/16 =** _____
5. **4 coins: P(exactly 3 Heads) = 4/16 =** _____
6. **5 coins: Ways to get 2 Heads (5C2) =** _____
7. **5 coins: P(exactly 2 Heads) = 10/32 =** _____
8. **Bag (3 Red, 2 Blue): P(2 Red) = 3C2/5C2 =** _____
9. **Bag (4 Red, 2 Blue): P(2 Red) = 4C2/6C2 =** _____
10. **Committee of 2 Boys (from 4) & 2 Girls (from 3) =** _____
11. **Circular arrangements of 5 people (4!) =** _____
12. **Circular arrangements of 6 people (5!) =** _____
13. **Variance of {2, 4, 6} (pop σ^2: ((4+0+4)/3)) =** _____
14. **Sample SD of {2, 5, 8} ($\sqrt{((9+0+9)/2)}$) =** _____
15. **Sample SD of {10, 14, 18} =** _____
16. **z-score: x = 85, μ = 75, σ = 5 →** _____
17. **z-score: x = 64, μ = 70, σ = 4 →** _____
18. **Expected value: 0.5(10) + 0.5(20) =** _____
19. **Expected value of 1 fair die (1..6) =** _____
20. **Expected value of 2 fair dice sum =** _____

Section 12: Base Conversions, Binary & Bitwise Logic

Binary, octal, and hexadecimal place value, bit shifting, and boolean gates. • Study the Champion Way & complete W1–W6 before starting Tests 12A–12D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 12: BASE CONVERSIONS, BINARY & BITWISE LOGIC

RULE 1 **Rule 1 — Group Binary into 4-Bit Nibbles (8-4-2-1) for Instant Hex Conversion: Every 4 binary bits map directly to 1 Hex digit (0000=0 .. 1010=A .. 1111=F). Example: 1010 1100_2 = AC_{16}!**

RULE 2 Rule 2 — Bitwise Shifts Are Powers of Two: Left shift x << k multiplies x by 2^k; right shift x >> k divides x by 2^k (integer part).

RULE 3 Rule 3 — Two's Complement & Bitwise NOT Shortcuts: In n-bit unsigned binary, NOT(x) = $(2^n - 1)$ - x, and 8-bit Two's Complement of -k is (256 - k) in Hex (so -1 = FF_{16}, -2 = FE_{16}).

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — 8-Bit Binary to Hexadecimal: Convert 10101100_2 to Hex Stack Keys: BIN 1 0 1 0 1 1 0 0 ↶ BASE HEX

SLOW / AMATEUR WAY (Avoid):

Slow Way: Convert to decimal (128 + 32 + 8 + 4 = 172), then divide by 16 to get 10 R 12.

CHAMPION SPEED WAY (Use This!):

Champion Way: Split into two 4-bit nibbles: 1010_2 = 10 = A and 1100_2 = 12 = C → AC_{16} instantly!

Archetype 2 — Bitwise AND / XOR Evaluation: Evaluate 12 AND 10 and 12 XOR 10 Stack Keys: 1 2 ENTER 1 0 AND

SLOW / AMATEUR WAY (Avoid):

Slow Way: Guess decimal operations.

CHAMPION SPEED WAY (Use This!):

Champion Way: Compare 4-bit 1100_2 (12) and 1010_2 (10): shared 1-bit is 1000_2 = 8 (AND); differing bits are 0110_2 = 6 (XOR)!

Archetype 3 — Alternating Bit Mask XOR: Evaluate 170 XOR 85 Stack Keys: 1 7 0 ENTER 8 5 XOR

SLOW / AMATEUR WAY (Avoid):

Slow Way: Convert 170 and 85 bit-by-bit on scratch paper.

CHAMPION SPEED WAY (Use This!):

Champion Way: 170 (10101010_2) and 85 (01010101_2) are exact bit complements adding to 255, so 170 XOR 85 = 255!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. **Hex $2F_{16}$ in decimal =** _____

Shortcut Cue: 2 × 16 + 15 = 32 + 15

W2. **12 (1100_2) XOR 10 (1010_2) =** _____

Shortcut Cue: Differing bits: 0110_2

W3. **Left shift: 7 << 3 =** _____

Shortcut Cue: $7 \times 2^3 = 7 \times 8$

W4. **Binary 10101100_2 in hexadecimal =** _____

Shortcut Cue: Nibbles 1010 (A) and 1100 (C)

W5. **Decimal 4095 (2^{12} - 1) in Hex =** _____

Shortcut Cue: Three 4-bit 1111 nibbles → FFF

W6. **8-bit Two's Complement of -1 (Hex) =** _____

Shortcut Cue: 256 - 1 = 255 = FF_{16}

Warm-Up Self-Check Answers: W1: 47 | W2: 6 | W3: 56 | W4: AC | W5: FFF | W6: FF

Section 12: Base Conversions, Binary & Bitwise Logic

Binary, octal, and hexadecimal place value, bit shifting, and boolean gates. • Foundation (Test 12A) & Core Fluency (Test 12B)

TEST 12A • LEVEL 1: FOUNDATION Goal: 4 min

4-Bit Binary & Decimal Conversions

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. **Binary 0001_2 in decimal =** _____
2. **Binary 0010_2 in decimal =** _____
3. **Binary 0011_2 in decimal =** _____
4. **Binary 0100_2 in decimal =** _____
5. **Binary 0101_2 in decimal =** _____
6. **Binary 0110_2 in decimal =** _____
7. **Binary 0111_2 in decimal =** _____
8. **Binary 1000_2 in decimal =** _____
9. **Binary 1001_2 in decimal =** _____
10. **Binary 1010_2 in decimal =** _____
11. **Binary 1100_2 in decimal =** _____
12. **Binary 1111_2 in decimal =** _____
13. **Decimal 5 in 4-bit binary =** _____
14. **Decimal 6 in 4-bit binary =** _____
15. **Decimal 9 in 4-bit binary =** _____
16. **Decimal 10 in 4-bit binary =** _____
17. **Decimal 12 in 4-bit binary =** _____
18. **Decimal 13 in 4-bit binary =** _____
19. **Decimal 14 in 4-bit binary =** _____
20. **Decimal 15 in 4-bit binary =** _____

TEST 12B • LEVEL 2: CORE FLUENCY Goal: 5 min

8-Bit Binary, Octal (Base 8) & Hexadecimal (Base 16)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. **Binary 10000_2 in decimal =** _____
2. **Binary 10100_2 in decimal =** _____
3. **Binary 11111_2 in decimal =** _____
4. **Binary 100000_2 in decimal =** _____
5. **Binary 1000000_2 in decimal =** _____
6. **Binary 11111111_2 in decimal =** _____
7. **Hex A_{16} in decimal =** _____
8. **Hex C_{16} in decimal =** _____
9. **Hex F_{16} in decimal =** _____
10. **Hex 10_{16} in decimal =** _____
11. **Hex $1A_{16}$ in decimal =** _____
12. **Hex 20_{16} in decimal =** _____
13. **Hex $2F_{16}$ in decimal =** _____
14. **Hex FF_{16} in decimal =** _____
15. **Decimal 16 in hexadecimal =** _____
16. **Decimal 31 in hexadecimal =** _____
17. **Decimal 64 in hexadecimal =** _____
18. **Decimal 255 in hexadecimal =** _____
19. **Decimal 8 in octal (Base 8) =** _____
20. **Decimal 64 in octal (Base 8) =** _____

Section 12: Base Conversions, Binary & Bitwise Logic

Binary, octal, and hexadecimal place value, bit shifting, and boolean gates. • Multi-Step Speed (Test 12C) & Advanced Challenge (Test 12D)

TEST 12C • LEVEL 3: MULTI-STEP SPEED Goal: 5 min

Bitwise Boolean Logic (AND, OR, XOR) & Bit Shifts

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. **1 AND 1 =** _____
2. **1 AND 0 =** _____
3. **1 OR 0 =** _____
4. **1 XOR 1 =** _____
5. **1 XOR 0 =** _____
6. **12 (1100_2) AND 10 (1010_2) =** _____
7. **12 (1100_2) OR 10 (1010_2) =** _____
8. **12 (1100_2) XOR 10 (1010_2) =** _____
9. **6 (0110_2) AND 3 (0011_2) =** _____
10. **6 (0110_2) OR 3 (0011_2) =** _____
11. **6 (0110_2) XOR 3 (0011_2) =** _____
12. **15 (1111_2) AND 9 (1001_2) =** _____
13. **15 (1111_2) XOR 9 (1001_2) =** _____
14. **255 AND 15 =** _____
15. **192 AND 240 =** _____
16. **Left shift: 5 << 1 (5 × 2) =** _____
17. **Left shift: 3 << 2 (3 × 4) =** _____
18. **Left shift: 7 << 3 (7 × 8) =** _____
19. **Right shift: 24 >> 2 (24 ÷ 4) =** _____
20. **Right shift: 64 >> 3 (64 ÷ 8) =** _____

TEST 12D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Mixed-Base Arithmetic, Two's Complement & Hex Masks

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. **Binary sum: $1010_2 + 0101_2$ (in dec) =** _____
2. **Binary sum: $1100_2 + 0110_2$ (in dec) =** _____
3. **Hex sum: $A_{16} + 6_{16}$ (in Hex) =** _____
4. **Hex sum: $F_{16} + 1_{16}$ (in Hex) =** _____
5. **Hex sum: $1A_{16} + 16_{16}$ (in Hex) =** _____
6. **Octal 17_8 in decimal =** _____
7. **Octal 77_8 in decimal =** _____
8. **Octal 100_8 in hexadecimal =** _____
9. **Binary 10101100_2 in hexadecimal =** _____
10. **Binary 11110000_2 in hexadecimal =** _____
11. **Hex $B4_{16}$ in 8-bit binary =** _____
12. **Decimal 4095 ($2^{12} - 1$) in Hex =** _____
13. **Max unsigned 8-bit integer ($2^8 - 1$) =** _____
14. **Max unsigned 10-bit integer ($2^{10} - 1$) =** _____
15. **4-bit bitwise NOT of 5 (15 - 5) =** _____
16. **8-bit bitwise NOT of 200 (255 - 200) =** _____
17. **8-bit Two's Complement of -1 (in Hex) =** _____
18. **8-bit Two's Complement of -2 (in Hex) =** _____
19. **(255 AND 240) >> 4 =** _____
20. **(170 XOR 85) (10101010_2 ^ 01010101_2) =** _____

Cumulative Review C3: All-Level Championship (Sections 1–12)

Grand championship speed tests spanning arithmetic, algebra, trig, logs, stats, and CS. • Study the Champion Way & complete W1–W6 before starting Tests C3A–C3D.

1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — CUMULATIVE REVIEW C3: ALL-LEVEL CHAMPIONSHIP (SECTIONS 1–12)

RULE 1 — **Rule 1 — Championship Triage: Solve 1-second identity problems (Vieta's roots, $\sin^2+\cos^2=1$, a^2-b^2, GCF×LCM, Pascal 2^n, nibble Hex) mentally, saving stack keystrokes for multi-step transcendentals.**

RULE 2 — Rule 2 — Zero Scratch-Paper Transcription on Transcendentals: Keep 64-bit precision on the stack from the deepest inner expression outward, rounding only the final display to 2 or 4 decimal places.

RULE 3 — Rule 3 — Use [→ϑ,r] for Every Hypotenuse, Distance, and Wave Amplitude Problem: 2D distance, vector magnitude, and max(a sin x + b cos x) are all solved by a single [→ϑ,r] operation.

2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

Archetype 1 — Mixed Fraction Quotient + Sum: Evaluate (3 1/3 ÷ 1 1/4) + 2/5 Stack Keys: [3] [.] [1] [.] [3] [ENTER] [1] [.] [1] [.] [4] [÷] [.] [2] [.] [5] [+]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Convert 10/3 ÷ 5/4 = 40/15 on paper, add 2/5, convert to decimal.

CHAMPION SPEED WAY (Use This!):

Champion Way: Key [3.1.3] [ENTER] [1.1.4] [÷] [.2.5] [+] → 3.0667 in 5 entries!

Archetype 2 — Continuous Compounding Sprint: Evaluate $1500 \times e^{0.05 \times 10}$ Stack Keys: [.] [0] [5] [ENTER] [1] [0] [×] [e^x] [1] [5] [0] [0] [×]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Compute 0.05 × 10, write 0.5, compute $e^{0.5}$, multiply by 1500.

CHAMPION SPEED WAY (Use This!):

Champion Way: Start at the exponent: [.05] [ENTER] [10] [×] [e^x] [1500] [×] → 2473.08!

Archetype 3 — Grand Championship Multi-Identity Check: Evaluate GCF(24, 36) × LCM(24, 36) Stack Keys: [2] [4] [ENTER] [3] [6] [×]

SLOW / AMATEUR WAY (Avoid):

Slow Way: Factor 24 and 36, find GCF = 12 and LCM = 72, multiply.

CHAMPION SPEED WAY (Use This!):

Champion Way: GCF(a, b) × LCM(a, b) = a × b = 24 × 36 = (30 - 6)(30 + 6) = 30^2 - 6^2 = 900 - 36 = 864!

3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

W1. **12 cos(45°) + 5 sin(45°) (4 dec) =** _____

Shortcut Cue: (12 + 5) × sin(45°) = 17 × 0.7071

W2. **Monthly: 250 × (1 + 0.06 ÷ 12)^{24} =** _____

Shortcut Cue: [.06][ENTER][12][÷][1][+][24][y^x][250][×]

W3. **1 + 2 + 3 + ... + 20 =** _____

Shortcut Cue: Gauss rule: 20 × 21 ÷ 2

W4. **GCF(24, 36) × LCM(24, 36) =** _____

Shortcut Cue: 24 × 36 = 30^2 - 6^2 = 900 - 36

W5. **max value of 5 sin(x) + 12 cos(x) =** _____

Shortcut Cue: Amplitude $\sqrt{(5^2 + 12^2)}$

W6. **5 coins: P(exactly 2 Heads) =** _____

Shortcut Cue: 5C2 ÷ 32 = 10 ÷ 32

Warm-Up Self-Check Answers: **W1: 12.0208** | **W2: 281.79** | **W3: 210** | **W4: 864** | **W5: 13** | **W6: 0.3125**

Cumulative Review C3: All-Level Championship (Sections 1–12)

Grand championship speed tests spanning arithmetic, algebra, trig, logs, stats, and CS. • Foundation (Test C3A) & Core Fluency (Test C3B)

TEST C3A • LEVEL 1: FOUNDATION Goal: 5 min

Core Cross-Topic Speed: Arithmetic, Fractions, Powers & Binary

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1. **(18 - 6) × 3 =** _____
2. **3/4 + 1/2 (decimal) =** _____
3. **GCF of 18 and 24 =** _____
4. **LCM of 6 and 8 =** _____
5. **Solve: 3x - 5 = 16** _____
6. **Hypotenuse for legs 5 and 12 =** _____
7. **sin(30°) + cos(60°) =** _____
8. **tan(45°) × 8 =** _____
9. **2^5 =** _____
10. **$\log_{10}(1000)$ =** _____
11. **$\ln(e^4)$ =** _____
12. **5! =** _____
13. **5C2 =** _____
14. **Mean of {4, 8, 12, 16} =** _____
15. **Binary 1010_2 in decimal =** _____
16. **Decimal 15 in 4-bit binary =** _____
17. **Hex 10_{16} in decimal =** _____
18. **12 AND 10 =** _____
19. **15^2 =** _____
20. **1000 - 375 =** _____

TEST C3B • LEVEL 2: CORE FLUENCY Goal: 5 min

Intermediate Championship: Quadratics, Vectors, Logs & Combinatorics

Date: _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1. **Sum of double 6 and near-double (8 + 9) =** _____
2. **((35 - 7) × 4) ÷ (14 + 2) =** _____
3. **(1/3 + 1/6) × 18 =** _____
4. **Greater root of $x^2 + 5x + 6 = 0$:** _____
5. **Positive root of $x^2 - 25 = 0$:** _____
6. **Slope from (-2, 1) to (2, 13) =** _____
7. **Vector <8, 15> → Magnitude r =** _____
8. **Hyp = 20, $\vartheta = 30°$ → Opp leg y =** _____
9. **2^{10} =** _____
10. **$\log_2(64)$ =** _____
11. **$1000 \times (1.05)^2$ =** _____
12. **Half-life (3 periods): $100 \times (0.5)^3$ =** _____
13. **6P3 =** _____
14. **7C3 =** _____
15. **8C3 =** _____
16. **Decimal 255 in hexadecimal =** _____
17. **Decimal 64 in octal (Base 8) =** _____
18. **12 OR 10 =** _____
19. **12 XOR 10 =** _____
20. **192 AND 240 =** _____

Cumulative Review C3: All-Level Championship (Sections 1–12)

Grand championship speed tests spanning arithmetic, algebra, trig, logs, stats, and CS. • Multi-Step Speed (Test C3C) & Advanced Challenge (Test C3D)

TEST C3C • LEVEL 3: MULTI-STEP SPEED Goal: 6 min

Advanced Championship: Multi-Step Rationals, Trig & Growth

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. **(3 1/3 ÷ 1 1/4) + 2/5 (4 dec) =** _____
2. **1 ÷ (1/4 + 1/6) (decimal) =** _____
3. **Distance from (-3, 5) to (5, -10) =** _____
4. **Evaluate $2x^2 + 4x - 5$ at $x = 3$:** _____
5. **Sum of roots of $x^2 - 11x + 28 = 0$:** _____
6. **$12 \cos(45°) + 5 \sin(45°)$ (4 dec) =** _____
7. **$\sin^2(25°) + \cos^2(25°)$ =** _____
8. **Dot product $\langle 3, 4\rangle \cdot \langle 5, -2\rangle$ =** _____
9. **Doubling time: ln(2) ÷ 0.07 (4 dec) =** _____
10. **Continuous: $500 \times e^{0.04 \times 5}$ (4 dec) =** _____
11. **Continuous: $1500 \times e^{0.05 \times 10}$ (2 dec) =** _____
12. **Monthly: $250 \times (1 + 0.06 \div 12)^{24}$ (2 dec) =** _____
13. **10C3 =** _____
14. **Arrangements of LEVEL (5! ÷ 4) =** _____
15. **2 dice: P(sum = 7) =** _____
16. **Decimal 4095 ($2^{12} - 1$) in Hex =** _____
17. **Hex $2F_{16}$ in decimal =** _____
18. **Left shift: 9 << 3 (9 × 8) =** _____
19. **24 × 26 =** _____
20. **35^2 =** _____

TEST C3D • LEVEL 4: ADVANCED CHALLENGE Goal: 6 min

Grand Invitational Speed Challenge

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. **1 + 2 + 3 + ... + 20 =** _____
2. **99 + 98 + 97 + 96 + 95 =** _____
3. **$25^2 - 15^2$ =** _____
4. **24 × (5/6 - 3/8) =** _____
5. **GCF(24, 36) × LCM(24, 36) =** _____
6. **Number of positive factors of 36 =** _____
7. **3D diagonal $\sqrt{(3^2 + 4^2 + 12^2)}$ =** _____
8. **Minimum value of $y = x^2 - 8x + 19$:** _____
9. **Solve for x: $2x + y = 15$, $x - y = 3$:** _____
10. **Law of Cosines: a=8, b=5, C=60° → c =** _____
11. **max value of $5 \sin(x) + 12 \cos(x)$ =** _____
12. **Change of base: ln(125) ÷ ln(5) =** _____
13. **$64^{2/3}$ =** _____
14. **Solve: $4^x = 32$ (decimal) =** _____
15. **Row 6 sum: 6C0 + 6C1 + ... + 6C6 =** _____
16. **5 coins: P(exactly 2 Heads) (decimal) =** _____
17. **Sample SD of {10, 14, 18} =** _____
18. **Binary 10101100_2 in hexadecimal =** _____
19. **8-bit Two's Complement of -1 (in Hex) =** _____
20. **(170 XOR 85) =** _____

Official Answer Key — Sections 1 to 3

Verified answers for Tests A, B, C & D (#1–#20) plus Optimal Champion Pipelines for Level 3 & 4 challenge items.

Section 1: Step Counting, Benchmarks & Sequences (Foundational to Advanced)

Test 1A (Level 1: Foundation)		Test 1B (Level 2: Core Fluency)		Test 1C (Level 3: Multi-Step Speed)		Test 1D (Level 4: Advanced Challenge)	
1. 4	11. 6	1. 6	11. 15	1. 36	11. 100	1. 15	11. 150
2. 6	12. 12	2. 8	12. 20	2. 48	12. 90	2. 30	12. 36
3. 9	13. 13	3. 10	13. 30	3. 43	13. 245	3. 25	13. 60
4. 5	14. 14	4. 15	14. 30	4. 51	14. 165	4. 30	14. 50
5. 9	15. 14	5. 30	15. 50	5. 70	15. 200	5. 40	15. 250
6. 6	16. 13	6. 16	16. 30	6. 55	16. 255	6. 75	16. 90
7. 9	17. 12	7. 14	17. 22	7. 125	17. 300	7. 28	17. 70
8. 7	18. 10	8. 25	18. 60	8. 80	18. 300	8. 52	18. 350
9. 4	19. 16	9. 15	19. 55	9. 135	19. 480	9. 90	19. 55
10. 7	20. 16	10. 35	20. 100	10. 100	20. 300	10. 56	20. 110

Champion Pipelines: • **1C #11 (68+12+20): Pair 68 + 12 = 80 first, then + 20 = 100** • **1D #3 (1+3+5+7+9): 5 odd terms → $5^2 = 25$**

• 1D #16 (14+16+18+20+22): 5 × middle term (18) = 90 • 1D #19 (1+2+...+10): 10 × 11 ÷ 2 = 55 ([10] [ENTER] [11] [×] [2] [÷])

Section 2: Number Bonds & Missing Addends (Foundational to Advanced)

Test 2A (Level 1: Foundation)		Test 2B (Level 2: Core Fluency)		Test 2C (Level 3: Multi-Step Speed)		Test 2D (Level 4: Advanced Challenge)	
1. 7	11. 3	1. 14	11. 8	1. 39	11. 33	1. 190	11. 40
2. 7	12. 11	2. 13	12. 9	2. 59	12. 43	2. 155	12. 35
3. 8	13. 11	3. 15	13. 9	3. 64	13. 34	3. 65	13. 185
4. 8	14. 11	4. 15	14. 9	4. 63	14. 56	4. 82	14. 320
5. 9	15. 12	5. 17	15. 11	5. 33	15. 83	5. 175	15. 200
6. 9	16. 12	6. 7	16. 7	6. 36	16. 95	6. 135	16. 275
7. 9	17. 12	7. 6	17. 14	7. 37	17. 37	7. 129	17. 265
8. 4	18. 6	8. 8	18. 7	8. 35	18. 52	8. 70	18. 180
9. 3	19. 5	9. 7	19. 15	9. 22	19. 45	9. 100	19. 275
10. 5	20. 4	10. 9	20. 9	10. 26	20. 46	10. 30	20. 400

Champion Pipelines: • **2C #17 (63 + ? = 100): 9-from-9, 10-from-10 rule → 37** • **2D #6 (48 + 35 + 52): Group (48 + 52) + 35 = 100 + 35 = 135**

• 2D #9 (200 - 68 - 32): 200 - (68 + 32) = 200 - 100 = 100 • 2D #11 (76 + 54 = 90 + ?): [76] [ENTER] [54] [+] [90] [-] → 40

Section 3: Complements & Make-10 Bridging (Foundational to Advanced)

Test 3A (Level 1: Foundation)		Test 3B (Level 2: Core Fluency)		Test 3C (Level 3: Multi-Step Speed)		Test 3D (Level 4: Advanced Challenge)	
1. 3	11. 19	1. 17	11. 34	1. 70	11. 66	1. 600	11. 57.5
2. 6	12. 13	2. 15	12. 37	2. 55	12. 84	2. 350	12. 146
3. 2	13. 15	3. 13	13. 42	3. 38	13. 67	3. 725	13. 163
4. 7	14. 14	4. 14	14. 24	4. 72	14. 93	4. 616	14. 283
5. 1	15. 12	5. 13	15. 32	5. 26	15. 64	5. 258	15. 455
6. 4	16. 13	6. 16	16. 36	6. 81	16. 63	6. 481	16. 735
7. 8	17. 15	7. 8	17. 23	7. 17	17. 66	7. 137	17. 297
8. 5	18. 12	8. 23	18. 55	8. 43	18. 125	8. 903	18. 291
9. 14	19. 14	9. 26	19. 67	9. 45	19. 139	9. 6.6	19. 600
10. 17	20. 16	10. 23	20. 47	10. 67	20. 136	10. 3.25	20. 3355

Champion Pipelines: • **3C #6 (100 - 19): 9-1=8, 10-9=1 → 81** • **3D #5 (1000 - 742): Left-to-right 9-9-10 rule → 258**

• 3D #16 (498 + 237): 500 + 235 = 735 • 3D #18 (98 + 97 + 96): 300 - (2 + 3 + 4) = 291

Official Answer Key — Sections 4 to 5

Verified answers for Tests A, B, C & D (#1–#20) plus Optimal Champion Pipelines for Level 3 & 4 challenge items.

Section 4: Doubles, Equal Groups & Squares (Foundational to Advanced)

Test 4A (Level 1: Foundation)		Test 4B (Level 2: Core Fluency)		Test 4C (Level 3: Multi-Step Speed)		Test 4D (Level 4: Advanced Challenge)	
1. 6	11. 13	1. 26	11. 12	1. 54	11. 96	1. 121	11. 99
2. 8	12. 15	2. 28	12. 20	2. 76	12. 120	2. 144	12. 143
3. 10	13. 17	3. 30	13. 18	3. 92	13. 200	3. 169	13. 224
4. 12	14. 19	4. 32	14. 30	4. 130	14. 37	4. 196	14. 399
5. 14	15. 22	5. 36	15. 32	5. 156	15. 48	5. 225	15. 624
6. 16	16. 24	6. 50	16. 42	6. 60	16. 65	6. 256	16. 899
7. 18	17. 21	7. 70	17. 12	7. 72	17. 21	7. 400	17. 450
8. 20	18. 23	8. 90	18. 18	8. 92	18. 30	8. 625	18. 244
9. 9	19. 12	9. 31	19. 25	9. 100	19. 78	9. 900	19. 200
10. 11	20. 18	10. 49	20. 24	10. 140	20. 50	10. 1225	20. 400

Champion Pipelines: • **4C #10 (4 × 35): Double 35 → 70, double 70 → 140** • **4D #10 (35^2): 3 × 4 = 12, append 25 → 1225**

• 4D #15 (24 × 26): 25^2 - 1 = 625 - 1 = 624 • 4D #20 (25^2 - 15^2): (25 + 15)(25 - 15) = 40 × 10 = 400

Cumulative Review C1: Number Sense (Sections 1–4) (Foundational to Advanced)

Test C1A (Level 1: Foundation)		Test C1B (Level 2: Core Fluency)		Test C1C (Level 3: Multi-Step Speed)		Test C1D (Level 4: Advanced Challenge)	
1. 8	11. 13	1. 27	11. 24	1. 63	11. 23	1. 21	11. 5.65
2. 10	12. 15	2. 20	12. 40	2. 32	12. 56	2. 42	12. 36.2
3. 9	13. 7	3. 9	13. 16	3. 84	13. 65	3. 535	13. 390
4. 5	14. 8	4. 8	14. 23	4. 36	14. 100	4. 218	14. 260
5. 2	15. 15	5. 16	15. 36	5. 76	15. 144	5. 446	15. 240
6. 7	16. 18	6. 32	16. 57	6. 55	16. 225	6. 563	16. 280
7. 16	17. 15	7. 32	17. 35	7. 96	17. 90	7. 196	17. 105
8. 14	18. 9	8. 48	18. 65	8. 76	18. 40	8. 625	18. 475
9. 16	19. 20	9. 35	19. 36	9. 120	19. 165	9. 399	19. 650
10. 13	20. 15	10. 21	20. 55	10. 55	20. 67	10. 899	20. 49

Champion Pipelines: • **C1C #16 (15^2): 1 × 2 = 2, append 25 → 225** • **C1D #10 (29 × 31): 30^2 - 1 = 899**

• C1D #14 (18^2 - 8^2): (18 + 8)(18 - 8) = 26 × 10 = 260 • C1D #20 (1+3+...+13): 7 odd numbers → 7^2 = 49

Section 5: Multi-Step Operations & Grouping (Grades 3–8+)

Test 5A (Level 1: Foundation)		Test 5B (Level 2: Core Fluency)		Test 5C (Level 3: Multi-Step Speed)		Test 5D (Level 4: Advanced Challenge)	
1. 24	11. 18	1. 36	11. 8	1. 69	11. 24	1. 110	11. 15
2. 35	12. 8	2. 5	12. 40	2. 28	12. 32	2. 108	12. 32
3. 48	13. 27	3. 5	13. 30	3. 4	13. 17	3. 8	13. 31
4. 63	14. 22	4. 5	14. 25	4. 10	14. 120	4. 10	14. 24
5. 8	15. 8	5. 36	15. 35	5. 72	15. 6	5. 100	15. 90
6. 9	16. 10	6. 8	16. 10	6. 72	16. 60	6. 10	16. 30
7. 9	17. 20	7. 40	17. 40	7. 90	17. 60	7. 100	17. 120
8. 7	18. 20	8. 90	18. 48	8. 28	18. 5	8. 96	18. 16
9. 18	19. 30	9. 80	19. 5	9. 8	19. 10	9. 600	19. 10
10. 5	20. 5	10. 10	20. 72	10. 56	20. 120	10. 5	20. 10

Champion Pipelines: • **5C #5 ((8+4)×(9-3)): [8] [ENTER] [4] [+] [9] [ENTER] [3] [-] [×] → 72** • **5C #12 (100-(16×5-12)): Inner 80 - 12 = 68 first, 100 - 68 = 32**

• 5D #15 ((28×9)-(18×9)): Distributive shortcut: (28 - 18) × 9 = 90 • 5D #16 ((45×12)÷18): Cancel common factors: 45 × (12/18) = 45 × 2/3 = 30

Official Answer Key — Sections 6 to 8

Verified answers for Tests A, B, C & D (#1–#20) plus Optimal Champion Pipelines for Level 3 & 4 challenge items.

Section 6: Fractions, Decimals & Reciprocals (Grades 4–9+)

Test 6A (Level 1: Foundation)		Test 6B (Level 2: Core Fluency)		Test 6C (Level 3: Multi-Step Speed)		Test 6D (Level 4: Advanced Challenge)	
1. 3/4	11. 0.25	1. 1/2	11. 1/2	1. 3.75	11. 10	1. 1	11. 2/5
2. 2/5	12. 0.75	2. 1/2	12. 12	2. 2.25	12. 0.875	2. 2	12. 1/5
3. 5/8	13. 0.2	3. 5/8	13. 4/3	3. 2.25	13. 0.125	3. 2	13. 2
4. 3/10	14. 0.6	4. 1/2	14. 1.25	4. 1.75	14. 0.375	4. 2.4	14. 2
5. 3/4	15. 2/5	5. 5/6	15. 2.5	5. 6	15. 0.625	5. 1	15. 16
6. 1/4	16. 1/2	6. 1/6	16. 2	6. 15	16. 0.875	6. 10	16. 5/3
7. 1	17. 8	7. 7/12	17. 3	7. 10	17. 3	7. 14	17. 7/4
8. 3/4	18. 5	8. 5/12	18. 1.5	8. 10	18. 3	8. 1	18. 2.5
9. 5/8	19. 9	9. 3/8	19. 1	9. 6	19. 1.5	9. 17	19. 11
10. 0.5	20. 10	10. 1/2	20. 12	10. 6	20. 10	10. 18	20. 19

Champion Pipelines: • **6C #12 (3/4 + 1/2×1/4): 0.75 + 0.125 = 0.875 ([.1.2] [ENTER] [.1.4] [×] [.3.4] [+])** • **6D #4 (1 ÷ (1/4 + 1/6)): Reciprocal pipeline: [4] [1/x] [6] [1/x] [+] [1/x] → 2.4**

• 6D #12 (1/2×2/3×3/4×4/5): Telescoping cancellation leaves 1/5 • 6D #19 (24 × (5/6 - 3/8)): Distribute 24: 20 - 9 = 11

Section 7: Primes, GCF/LCM & Nested Expressions (Grades 5–10+)

Test 7A (Level 1: Foundation)		Test 7B (Level 2: Core Fluency)		Test 7C (Level 3: Multi-Step Speed)		Test 7D (Level 4: Advanced Challenge)	
1. 12	11. 125	1. 12	11. 7	1. 50	11. 50	1. 216	11. 28
2. 18	12. 2	2. 15	12. 4	2. 30	12. 5	2. 300	12. 10
3. 20	13. 17	3. 16	13. 6	3. 20	13. 40	3. 36	13. 10
4. 24	14. 4	4. 14	14. 10	4. 12	14. 24	4. 24	14. 40
5. 30	15. 5	5. 24	15. 72	5. 40	15. 4	5. 36	15. 7
6. 36	16. 6	6. 36	16. 60	6. 5	16. 10	6. 120	16. 4
7. 8	17. 12	7. 60	17. 200	7. 70	17. 60	7. 6	17. 5
8. 16	18. 12	8. 70	18. 25	8. 8	18. 8	8. 8	18. 10
9. 27	19. 24	9. 20	19. 16	9. 36	19. 5	9. 9	19. 60
10. 64	20. 10	10. 10	20. 36	10. 30	20. 125	10. 10	20. 13

Champion Pipelines: • **7D #2 (GCF×LCM of 15,20): Identity a × b = 15 × 20 = 300** • **7D #9 (Factors of 36): $36 = 2^2 \times 3^2 \rightarrow (2+1)(2+1) = 9$ factors**

• 7D #14 ((25²-15²)÷(25-15)): $(a^2 - b^2)/(a - b) = a + b = 25 + 15 = 40$ • 7D #20 ((12²+5²)÷13): Pythagorean $5^2 + 12^2 = 13^2$, so $13^2 \div 13 = 13$

Section 8: Algebra, Quadratics & Coordinate Geometry (Grades 6–10+)

Test 8A (Level 1: Foundation)		Test 8B (Level 2: Core Fluency)		Test 8C (Level 3: Multi-Step Speed)		Test 8D (Level 4: Advanced Challenge)	
1. 17	11. 6	1. 3	11. 13	1. 20	11. 3	1. 9	11. 7
2. 13	12. 7	2. 2	12. 17	2. 25	12. 5	2. 20	12. 3
3. 18	13. 5	3. 3	13. 15	3. 4	13. 3	3. -11	13. 5
4. 12	14. 5	4. 3	14. 20	4. 17	14. 4	4. 24	14. 7
5. 14	15. 12	5. -2	15. 8	5. -2	15. 5	5. 5	15. 5
6. 12	16. 7	6. -0.5	16. 12	6. -3	16. -5	6. 3.5	16. 25
7. 23	17. -8	7. 5	17. 5	7. -3	17. 4	7. 16	17. 41
8. 9	18. 24	8. 7	18. 5	8. -3	18. -0.5	8. 0	18. 13
9. 35	19. -4	9. 5	19. 10	9. 3	19. 3	9. 4	19. 7
10. 6	20. 25	10. 10	20. 17	10. 2	20. 5	10. 3	20. 9

Champion Pipelines: • **8C #5 (Roots of $x^2+5x+6=0$): [1] [ENTER] [5] [ENTER] [6] [POLY] → -2, -3** • **8D #5 (Sum of roots $2x^2-10x+7$): Vieta's $-b/a = 10 / 2 = 5$**

• 8D #10 (Min of $x^2-8x+19$): Complete square: $(x - 4)^2 + 3 \rightarrow$ min is 3 • 8D #18 ($\sqrt{(3^2+4^2+12^2)}$): Chained (3,4,5) + (5,12,13) → 13

Official Answer Key — Sections C2 to 10

Verified answers for Tests A, B, C & D (#1–#20) plus Optimal Champion Pipelines for Level 3 & 4 challenge items.

Cumulative Review C2: Intermediate Math (Sections 1–8) (Grades 5–10+)

Test C2A (Level 1: Foundation)		Test C2B (Level 2: Core Fluency)		Test C2C (Level 3: Multi-Step Speed)		Test C2D (Level 4: Advanced Challenge)	
1. 27	11. 12	1. 1.75	11. 3	1. 0.875	11. -5	1. 55	11. 17
2. 64	12. 20	2. 3.75	12. 20	2. 2.4	12. 7	2. 36	12. 13
3. 36	13. 10	3. 1.25	13. 10	3. 14	13. 12	3. 1225	13. 9
4. 5	14. 18	4. 1.5	14. 3	4. 17	14. 7	4. 400	14. 5
5. 0.75	15. 14	5. 15	15. 4	5. 12	15. 12	5. 11	15. 7
6. 0.25	16. 7	6. 10	16. -2	6. 40	16. 15	6. 5/3	16. 25
7. 12	17. 14	7. 32	17. 2	7. 70	17. 8	7. 300	17. 30
8. 18	18. 5	8. 50	18. 28	8. 25	18. 24	8. 28	18. 616
9. 24	19. 10	9. 12	19. 225	9. 3	19. 399	9. 7	19. 294
10. 6	20. 4	10. 24	20. 625	10. -0.5	20. 624	10. 13	20. 1/3

Champion Pipelines: • **C2C #2 (1 ÷ (1/4 + 1/6)): [4] [1/x] [6] [1/x] [+] [1/x] → 2.4** • **C2D #5 (24 × (5/6 - 3/8)): Distribute 24: 20 - 9 = 11**

• C2D #11 (Dist (-3,5) to (5,-10)): dx = 8, dy = 15 → (8,15,17) triple = 17 • C2D #14 (Min of x^2-6x+14): Vertex at x = 3 → 9 - 18 + 14 = 5

Section 9: Trigonometry, Vectors & Polar Coordinates (Grades 8–12+)

Test 9A (Level 1: Foundation)		Test 9B (Level 2: Core Fluency)		Test 9C (Level 3: Multi-Step Speed)		Test 9D (Level 4: Advanced Challenge)	
1. 0	11. 1	1. 5	11. 7	1. 0.25	11. 5	1. 12.0208	11. 20
2. 1	12. 5	2. 6	12. 9	2. 0.75	12. 10	2. 8.6603	12. 0.5
3. 0.5	13. 30	3. 10	13. 5	3. 0.5	13. 13	3. 7.0711	13. 0.8660
4. 0.5	14. 60	4. 8	14. 10	4. 0.5	14. 23	4. 5.7735	14. 2
5. 1	15. 45	5. 9	15. 13	5. 3	15. 0	5. 53.13	15. 1
6. 0	16. 55	6. 14	16. 10	6. 1	16. 20	6. 36.87	16. 5
7. 0	17. 28	7. 10	17. 15	7. 5	17. 18	7. 45	17. 13
8. 1	18. π	8. 10	18. 17	8. 3	18. 10	8. -90	18. 17
9. 1	19. π/2	9. 0	19. 20	9. 4	19. 4	9. 19	19. 180
10. 2	20. π/3	10. 8	20. 25	10. 0.25	20. 2	10. 7	20. 90

Champion Pipelines: • **9D #1 (12 cos 45° + 5 sin 45°): Factor (12 + 5) × sin(45°): [17] [ENTER] [45] [SIN] [×] → 12.0208** • **9D #5 (Angle of <3, 4>): [3] [ENTER] [4] [yellow] [→θ,r] [x swap y] → 53.13°**

• 9D #10 (Law of Cosines a=8,b=5,C=60°): cos(60°)=0.5 cancels the 2 in 2ab cos C: $c^2 = a^2+b^2-ab = 64+25-40 = 49 \rightarrow 7$ • 9D #17 (max of 5 sin x + 12 cos x): Amplitude $\sqrt{(5^2 + 12^2)} = 13$

Section 10: Exponents, Logarithms & Growth Models (Grades 8–12+)

Test 10A (Level 1: Foundation)		Test 10B (Level 2: Core Fluency)		Test 10C (Level 3: Multi-Step Speed)		Test 10D (Level 4: Advanced Challenge)	
1. 1	11. 625	1. 0	11. -2	1. 2	11. 32	1. 9.9021	11. 3
2. 1	12. 1000	2. 1	12. 5	2. 3	12. 50	2. 13.8629	12. 2
3. 16	13. 10000	3. 2	13. 9	3. 2	13. 25	3. 610.7014	13. 1.5
4. 32	14. 0.5	4. 3	14. 7	4. 3	14. 12.5	4. 2473.08	14. 1.5
5. 64	15. 0.25	5. 4	15. 3	5. 3	15. 10	5. 281.79	15. 9
6. 256	16. 0.2	6. -1	16. 5	6. 2	16. 400	6. 1331	16. 8
7. 1024	17. 0.01	7. -2	17. 6	7. 6	17. 180	7. 3	17. 8
8. 27	18. 4	8. 0	18. 3	8. 4	18. 1100	8. 4	18. 16
9. 81	19. 3	9. 1	19. 3	9. 3	19. 1210	9. 3	19. 2.5
10. 125	20. 4	10. 3	20. 1	10. 1000	20. 1102.5	10. 5	20. 1.5

Champion Pipelines: • **10D #1 (ln(2) ÷ 0.07): [2] [LN] [.07] [÷] → 9.9021** • **10D #3 (500 × $e^{0.04\times5}$): [.04] [ENTER] [5] [×] [e^x] [500] [×] → 610.7014**

• 10D #5 (250 × $(1+0.06/12)^{24}$): [.06] [ENTER] [12] [÷] [1] [+] [24] [y^x] [250] [×] → 281.79 • 10D #18 ($64^{2/3}$): $(\sqrt[3]{64})^2 = 4^2 = 16$

Official Answer Key — Sections 11 to C3

Verified answers for Tests A, B, C & D (#1–#20) plus Optimal Champion Pipelines for Level 3 & 4 challenge items.

Section 11: Combinatorics, Probability & Statistics (Grades 7–12+)

Test 11A (Level 1: Foundation)		Test 11B (Level 2: Core Fluency)		Test 11C (Level 3: Multi-Step Speed)		Test 11D (Level 4: Advanced Challenge)	
1. 1	11. 8	1. 5	11. 20	1. 0.25	11. 120	1. 16	11. 24
2. 1	12. 9	2. 6	12. 21	2. 0.125	12. 3	2. 32	12. 120
3. 2	13. 8	3. 20	13. 35	3. 0.875	13. 12	3. 64	13. 8/3
4. 6	14. 18	4. 30	14. 28	4. 36	14. 30	4. 0.375	14. 3
5. 24	15. 5	5. 120	15. 56	5. 6	15. 15	5. 0.25	15. 4
6. 120	16. 0.5	6. 42	16. 36	6. 1/6	16. 28	6. 10	16. 2
7. 720	17. 1/6	7. 6	17. 45	7. 1/36	17. 9	7. 0.3125	17. -1.5
8. 15	18. 0.5	8. 10	18. 120	8. 1/36	18. 60	8. 0.3	18. 15
9. 6	19. 1/3	9. 10	19. 66	9. 6	19. 16	9. 0.4	19. 3.5
10. 20	20. 12	10. 15	20. 100	10. 24	20. 0.0667	10. 18	20. 7

Champion Pipelines: • **11B #15 (8C3): $(8 \times 7 \times 6) \div 6 = 56$** • **11C #14 (Arrangements of LEVEL): $5! \div (2! \times 2!) = 120 \div 4 = 30$**

• 11D #3 (Row 6 sum 6C0+...+6C6): $2^6 = 64$ • 11D #15 (Sample SD of {10,14,18}): 3 evenly spaced values with step d = 4 → SD = 4

Section 12: Base Conversions, Binary & Bitwise Logic (Grades 7–12+)

Test 12A (Level 1: Foundation)		Test 12B (Level 2: Core Fluency)		Test 12C (Level 3: Multi-Step Speed)		Test 12D (Level 4: Advanced Challenge)	
1. 1	11. 12	1. 16	11. 26	1. 1	11. 5	1. 15	11. 10110100
2. 2	12. 15	2. 20	12. 32	2. 0	12. 9	2. 18	12. FFF
3. 3	13. 0101	3. 31	13. 47	3. 1	13. 6	3. 10	13. 255
4. 4	14. 0110	4. 32	14. 255	4. 0	14. 15	4. 10	14. 1023
5. 5	15. 1001	5. 64	15. 10	5. 1	15. 192	5. 30	15. 10
6. 6	16. 1010	6. 255	16. 1F	6. 8	16. 10	6. 15	16. 55
7. 7	17. 1100	7. 10	17. 40	7. 14	17. 12	7. 63	17. FF
8. 8	18. 1101	8. 12	18. FF	8. 6	18. 56	8. 40	18. FE
9. 9	19. 1110	9. 15	19. 10	9. 2	19. 6	9. AC	19. 15
10. 10	20. 1111	10. 16	20. 100	10. 7	20. 8	10. F0	20. 255

Champion Pipelines: • **12C #15 (192 AND 240): $192 = 128 + 64$, both inside top mask 240 → 192** • **12D #9 (10101100_2 in Hex): $1010_2 = A$, $1100_2 = C$ → AC**

• 12D #12 (4095 in Hex): $2^{12} - 1 = 12$ ones = three F nibbles → FFF • 12D #20 (170 XOR 85): Complementary bit masks 10101010_2 ^ $01010101_2 = 11111111_2 = 255$

Cumulative Review C3: All-Level Championship (Sections 1–12) (Grades 7–12+)

Test C3A (Level 1: Foundation)		Test C3B (Level 2: Core Fluency)		Test C3C (Level 3: Multi-Step Speed)		Test C3D (Level 4: Advanced Challenge)	
1. 36	11. 4	1. 29	11. 1102.5	1. 3.0667	11. 2473.08	1. 210	11. 13
2. 1.25	12. 120	2. 7	12. 12.5	2. 2.4	12. 281.79	2. 485	12. 3
3. 6	13. 10	3. 9	13. 120	3. 17	13. 120	3. 400	13. 16
4. 24	14. 10	4. -2	14. 35	4. 25	14. 30	4. 11	14. 2.5
5. 7	15. 10	5. 5	15. 56	5. 11	15. 1/6	5. 864	15. 64
6. 13	16. 1111	6. 3	16. FF	6. 12.0208	16. FFF	6. 9	16. 0.3125
7. 1	17. 16	7. 17	17. 100	7. 1	17. 47	7. 13	17. 4
8. 8	18. 8	8. 10	18. 14	8. 7	18. 72	8. 3	18. AC
9. 32	19. 225	9. 1024	19. 6	9. 9.9021	19. 624	9. 6	19. FF
10. 3	20. 625	10. 6	20. 192	10. 610.7014	20. 1225	10. 7	20. 255

Champion Pipelines: • **C3C #1 ((3 1/3 ÷ 1 1/4) + 2/5): [3.1.3] [ENTER] [1.1.4] [÷] [.2.5] [+] → 3.0667** • **C3C #6 ($12 \cos 45° + 5 \sin 45°$): $17 \times \sin(45°) \rightarrow 12.0208$**

• C3D #5 (GCF(24,36)×LCM(24,36)): $24 \times 36 = (30 - 6)(30 + 6) = 900 - 36 = 864$ • C3D #10 (Law of Cosines a=8,b=5,C=60°): $c^2 = 64 + 25 - 40 = 49 \rightarrow c = 7$

Inside-Out Stack & Speed Reference Sheet

Five zero-memory RPN stack patterns and eight essential algebraic/geometry identities for speed competitions.

1. DEEPEST INNER EXPRESSION FIRST (ZERO PARENTHESES)

Evaluate the deepest nested group first so intermediate results wait in Register Y. Example: 100 ÷ (3 + 4×5):

Hardware Pipeline: [4] [ENTER] [5] [×] [3] [+] [1] [0] [0] [↑↓] [÷]

2. INSTANT DOUBLING & SQUARING ([ENTER] [+] AND [ENTER] [×])

Pressing [ENTER] duplicates Register X into Register Y without retyping digits. To double 18 and add 5:

Hardware Pipeline: [1] [8] [ENTER] [+] [5] [+]

3. NON-COMMUTATIVE SWAP (X SWAP Y) & RECIPROCAL ([1/x])

When denominator D is computed first, type numerator N and press [x swap y] [÷]—or invert D with [1/x] and multiply:

Hardware Pipeline: [3] [ENTER] [6] [+] [x^{-1}] [4] [5] [×]

4. PARALLEL RESISTANCE & HARMONIC SUMS (AUTOMATIC STACK LIFT)

For 1 / (1/4 + 1/6), automatic stack lift pushes the first reciprocal into Y while the second is computed:

Hardware Pipeline: [4] [x^{-1}] [6] [x^{-1}] [+] [x^{-1}]

5. 2D VECTOR MAGNITUDE, HYPOTENUSE & WAVE AMPLITUDE ([→ϑ,r])

Find the hypotenuse $\sqrt{(5^2 + 12^2)}$ or wave amplitude max(a sin x + b cos x) in one operation on the SC-32:

Hardware Pipeline: [1] [2] [ENTER] [5] [↶] [▸ϑ,r]

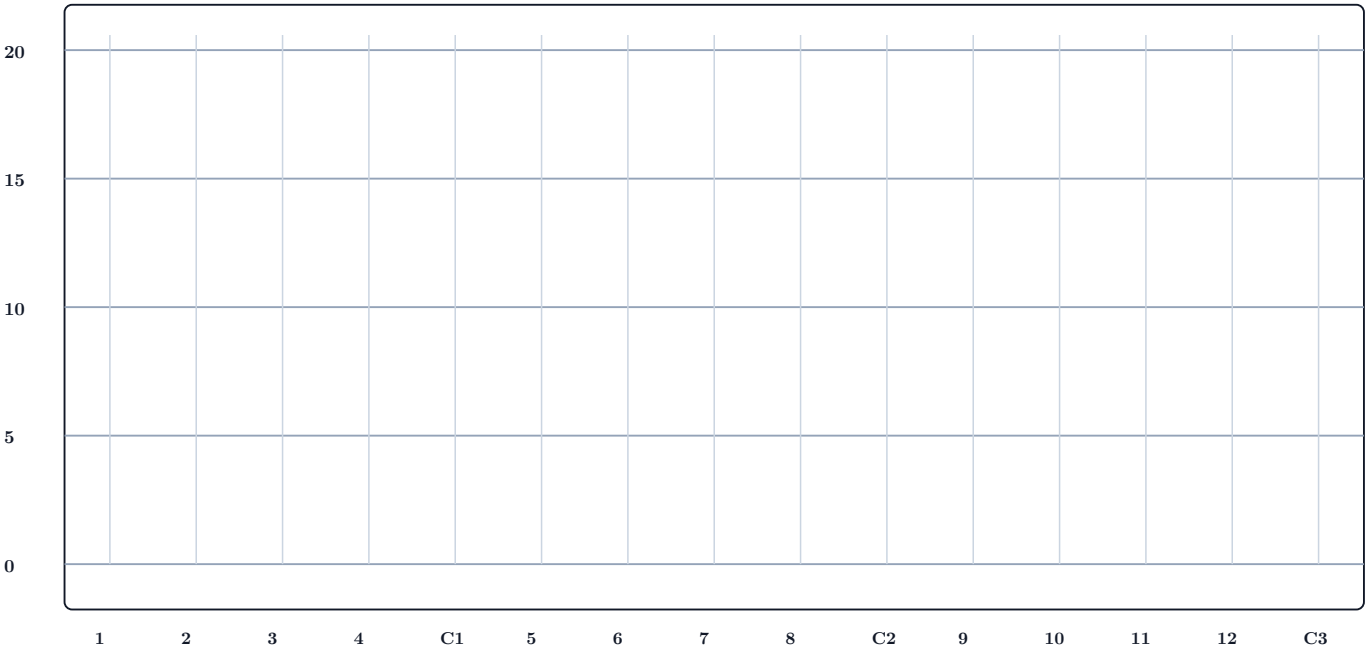
CHAMPIONSHIP ALGEBRAIC, NUMBER SENSE & GEOMETRY SPEED IDENTITIES

Gauss Series & Odd Sum	$1 + 2 + ... + n = n(n + 1) \div 2$ | Sum of first k odd numbers $(1 + 3 + 5 + ...) = k^2$
Sum of Squares & Cubes	$1^2 + 2^2 + ... + n^2 = n(n + 1)(2n + 1) \div 6$ | $1^3 + 2^3 + ... + n^3 = [n(n + 1) \div 2]^2$
Difference of Squares	$a^2 - b^2 = (a - b)(a + b)$ | Symmetric Product: $(m - d)(m + d) = m^2 - d^2$ (e.g., $28 \times 32 = 30^2 - 2^2 = 896$)
Squares Ending in 5	$(10t + 5)^2 = 100 \times t(t + 1) + 25$ (e.g., 35^2: $3 \times 4 = 12 \rightarrow 1225$; 75^2: $7 \times 8 = 56 \rightarrow 5625$)
GCF × LCM & Divisors	GCF(a, b) × LCM(a, b) = a × b | Number of divisors of $p^a \times q^b \times r^c = (a + 1)(b + 1)(c + 1)$
Vieta's Quadratic Rules	For $ax^2 + bx + c = 0$: Sum of Roots $(r_1 + r_2) = -b / a$ | Product of Roots $(r_1 \times r_2) = c / a$
2×2 Sum/Difference System	If x + y = S and x - y = D, then x = (S + D) ÷ 2 (Average) and y = (S - D) ÷ 2 (Half-Difference)
3D Diagonal & Equilateral	3D Box Diagonal $d = \sqrt{(l^2 + w^2 + h^2)}$ | Equilateral Triangle Area $A = (\sqrt{3} \div 4)s^2 \approx 0.43301 \times s^2$

Speed & Fluency Progress Graph & Certificate

Plot your highest score (out of 20) across all 15 sections and record your level completion.

1. 15-SECTION SCORE PROGRESSION GRAPH (PLOT YOUR SCORE FROM 0 TO 20)



CERTIFICATE OF MATHEMATICAL SPEED & FLUENCY

This certifies that: _____

has completed the speed lessons, guided warm-up drills, and progressive practice test series in Stack Speed & Fluency, demonstrating mathematical accuracy, multi-step structural fluency, and confident verification across Levels 1 through 4.

[ ] Level 1: Foundation Mastery (Tests A) **[ ] Level 2: Core Fluency Mastery (Tests B)**

[ ] Level 3: Multi-Step Speed (Tests C) **[ ] Level 4: Advanced Challenge (Tests D)**

Teacher / Parent / Coach Signature: _____ Date: _____